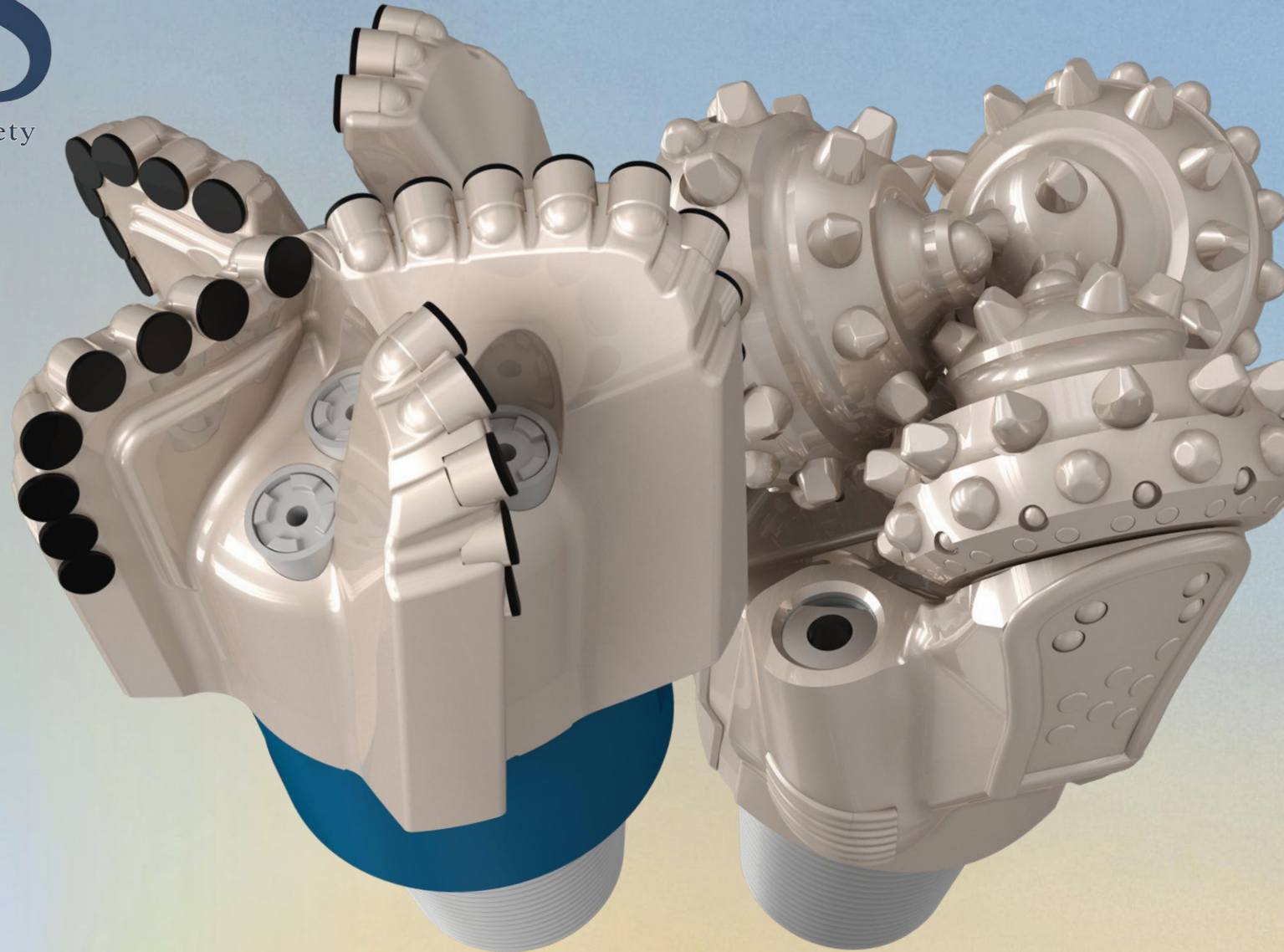




Drill Bit Introduction

17th July 2025

Rahmad Febdiansyah Bas



Agenda

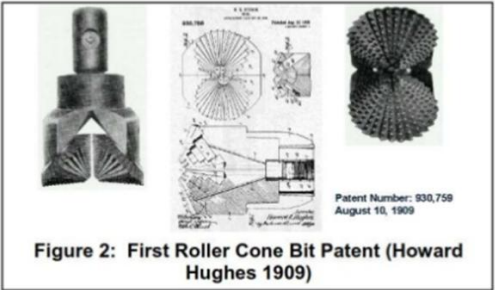
- 05:00 – 06:00** **Introduction to Drilling Bits**
Roller-Cone Bits - Fixed Cutter Bits
- 06:45 – 08:00** **Bit Hydraulics & Cleaning Efficiency**
Bit Selection Criteria
Bit Performance & Failure Analysis
- 08:00 – end** **Kahoot Quiz**



Drill Bit History

Howard Hughes
Patented the roller cone rock bit.

1925 – Intermeshing 2-cone bits invented



1909

1859

Edwin L. Drake
Drilled the first oilwell in Titusville, Pennsylvania, US

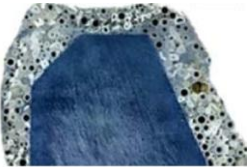
The first bit used was Fishtail Bit or Drag Bit.



1928-1932

1928 - Use of **tungsten carbide hardfacing** on bits begins

1932 - **Roller and ball bearings** introduced into roller cone bits



Three cone bits invented



1933

1953

General Electric GE create synthetic diamond crystals

Tungsten Carbide Inserts (TCI) first used in Roller Cone Bits.



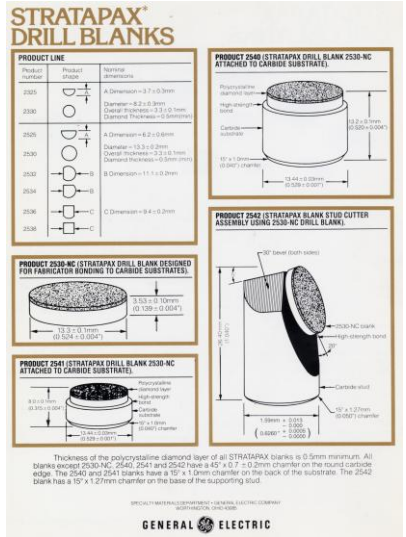
1951

1963

Sealed bearing roller cone bits first used

1973 - GE test the first 8mm Polycrystalline Diamond Compact (PDC).

1976-GE commercialized Stratapax Product line of PDC cutters.



Drill Bit History

Three cone bits invented

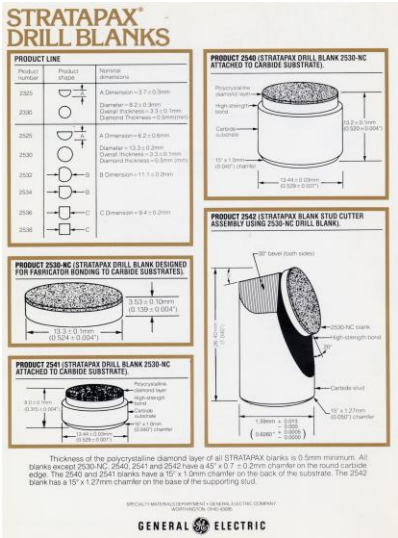
Tungsten Carbide Inserts (TCI)
first used in Roller Cone Bits.



1933



1951

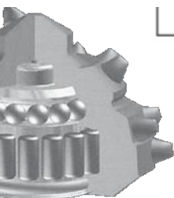


Non-planar interface NPI
technology introduced

1994

32

Tungsten carbide
begins
all bearings
roller cone bits



General Electric GE create
synthetic diamond crystals

1953

Sealed bearing
roller cone bits
first used

1963

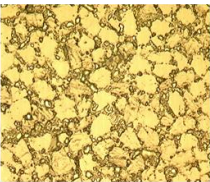
1973 - GE test the first
8mm Polycrystalline
Diamond Compact (PDC).

1973-1976

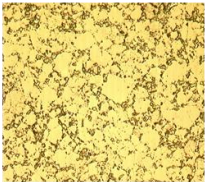
1976-GE commercialized
Stratapax Product line of
PDC cutters.

Mono and multi modal
grain mixes introduced

1990



MonoModal x 700

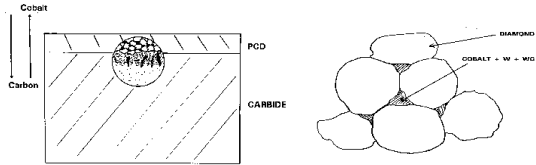


MultiModal x 700

Drill Bit History

TReX™ leaching technology
introduced by ReedHycalog

TReX®



2003



First combined PDC and
RC drill bit
commercialized
2009



Shaped PDC
technology re-
introduction takes off
2015

1996

Fully diamond
enhanced inserts first
used in roller cone
bits.



2005

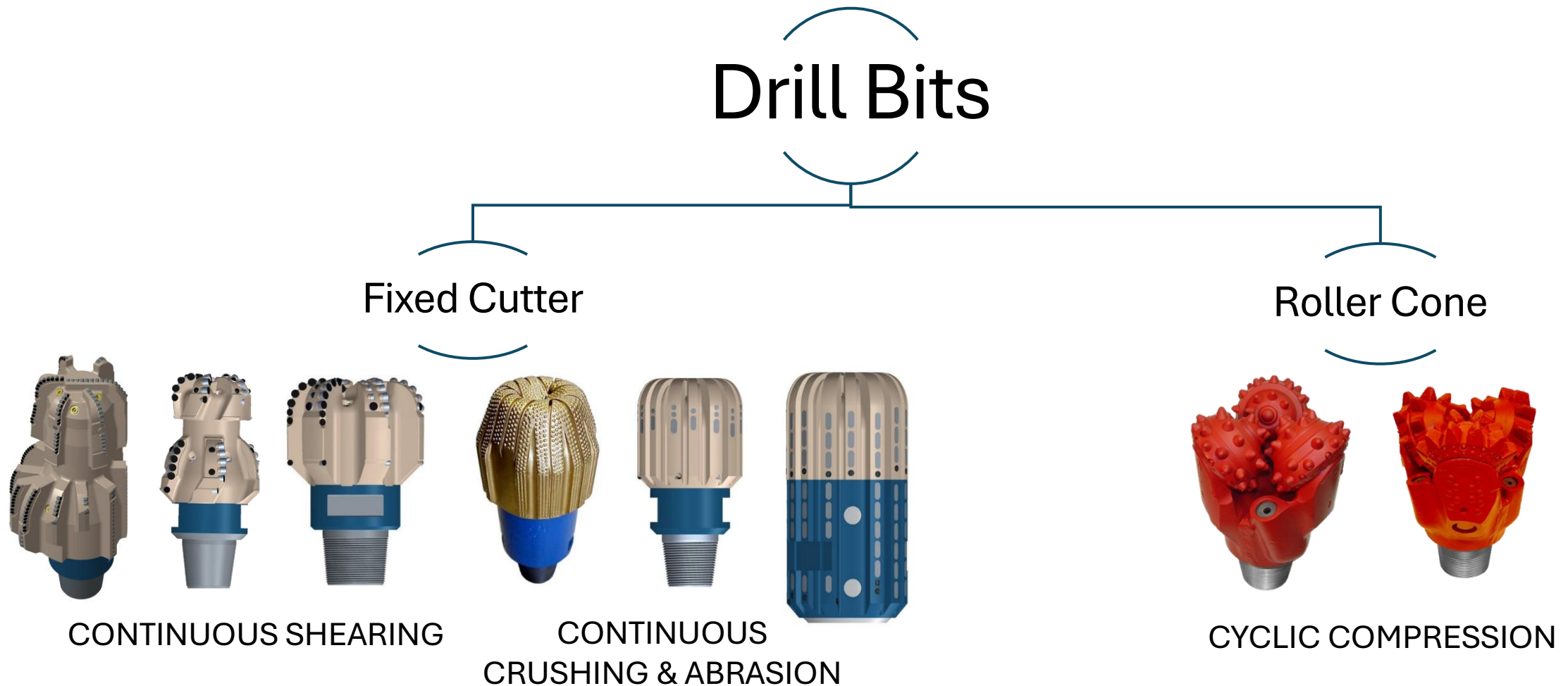
Fixed cutter bit market
size overtakes roller
cone

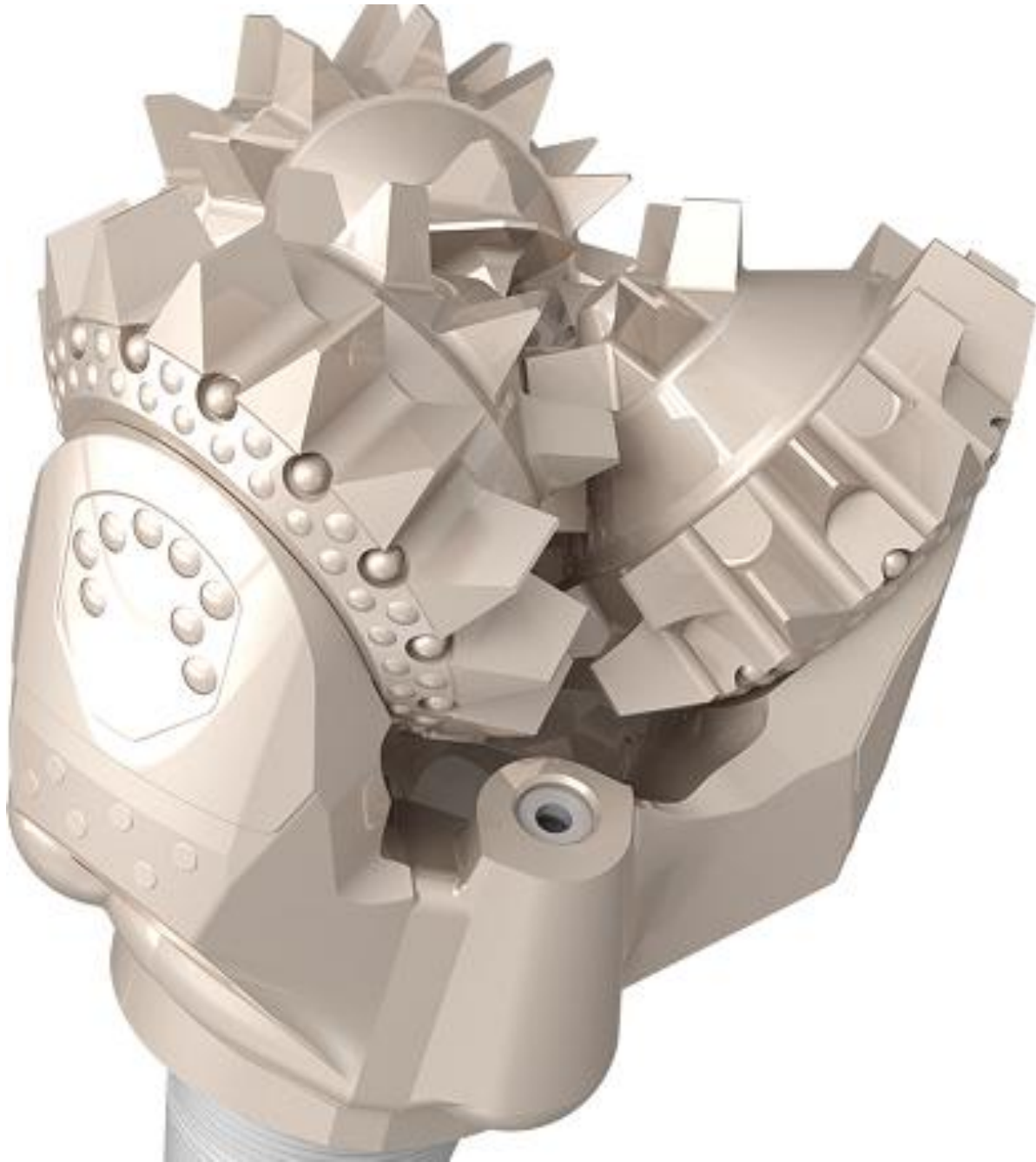
2013

Rotating PDC cutters
introduced



Drill Bit Types

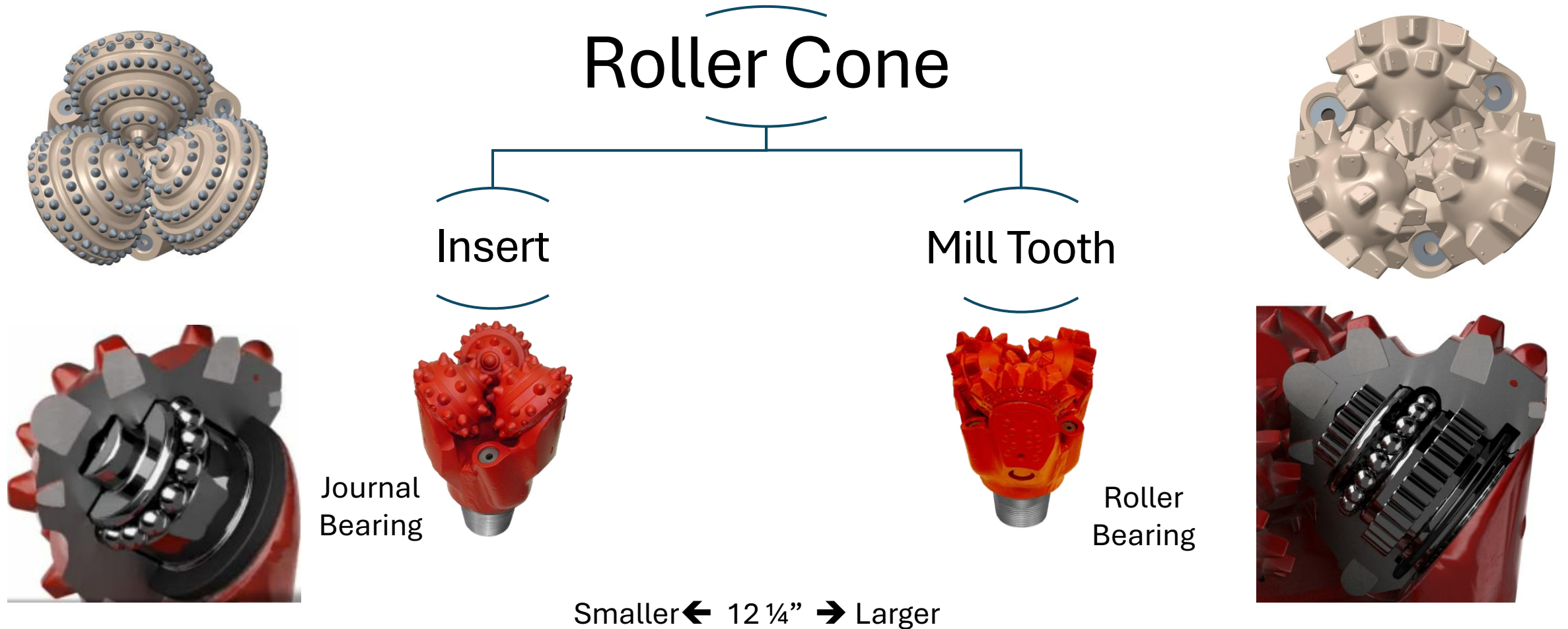




Roller Cones

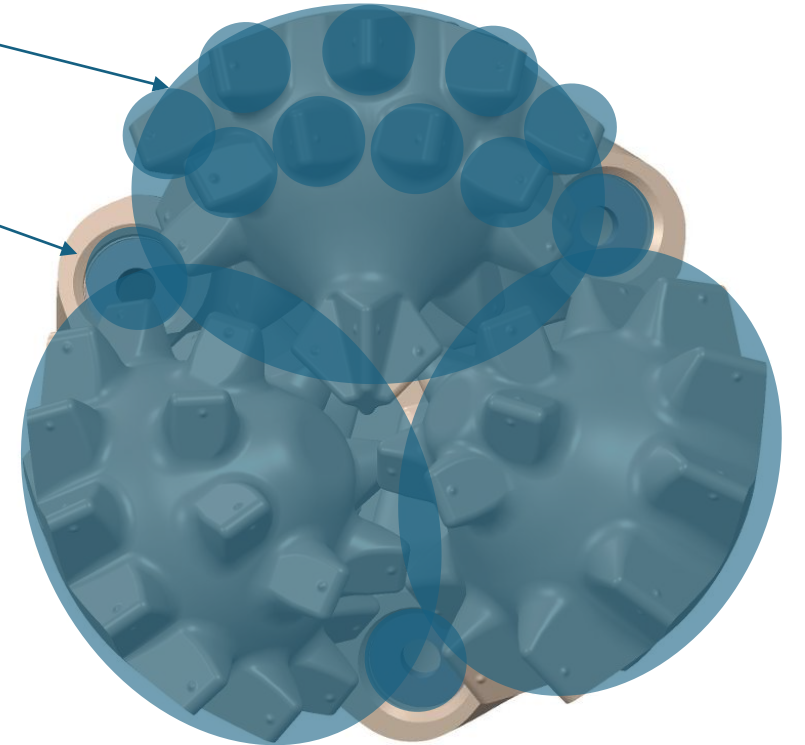
Drill Bit Types

Roller Cone



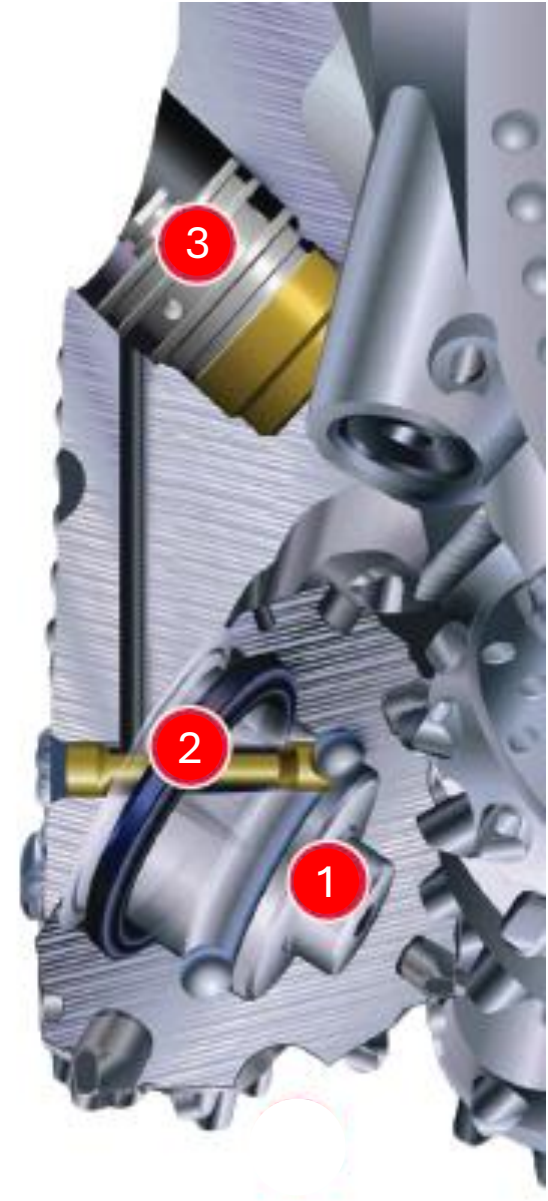
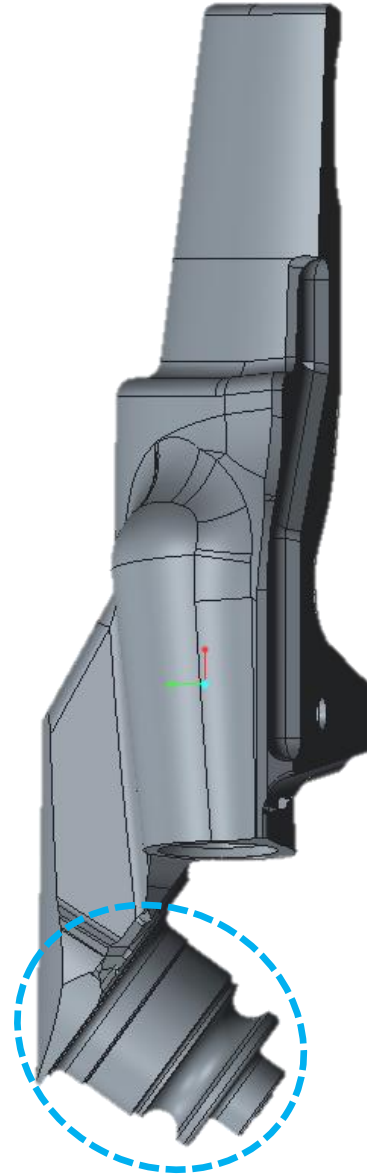
Basic Components of RC bit

- Cone
- Tooth
- Nozzle
- Arm
- Connection

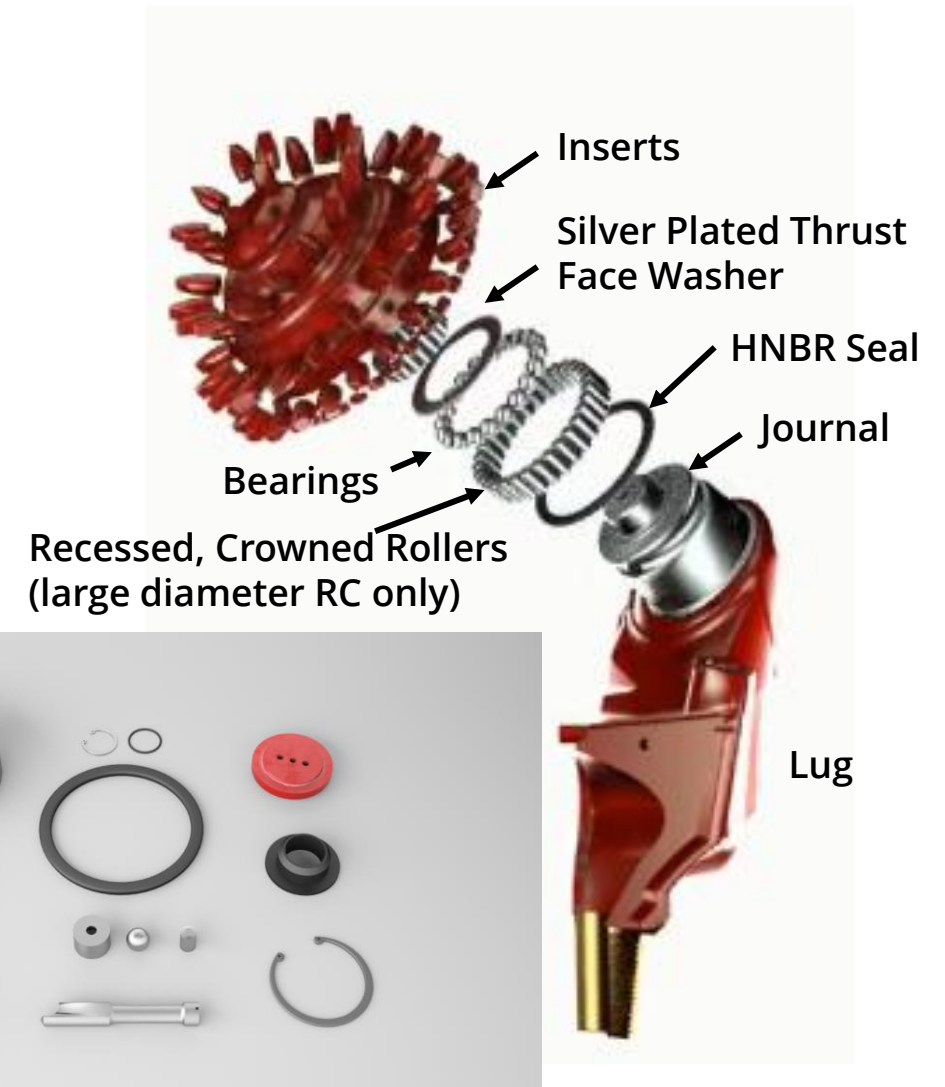
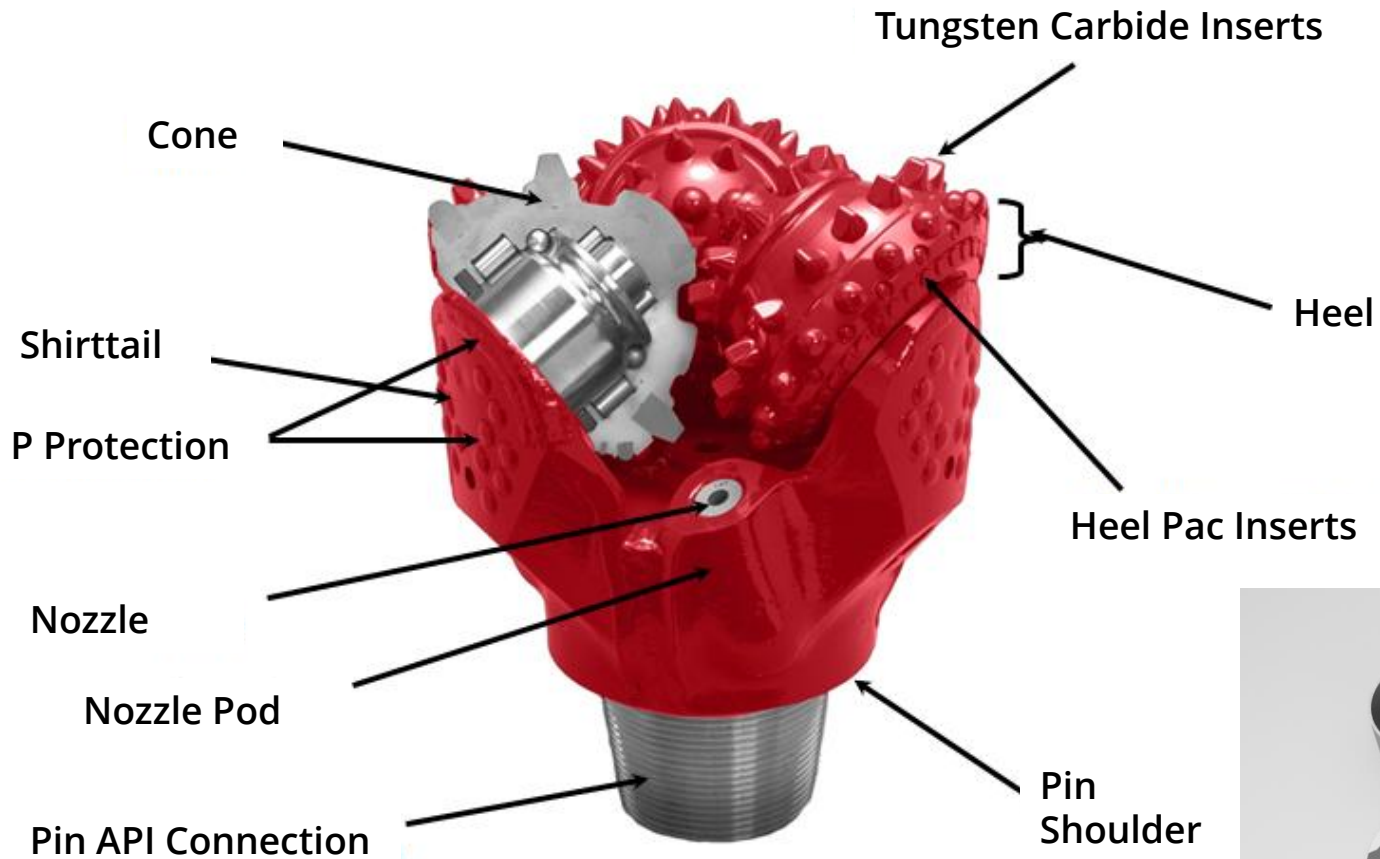


Basic Components of RC bit

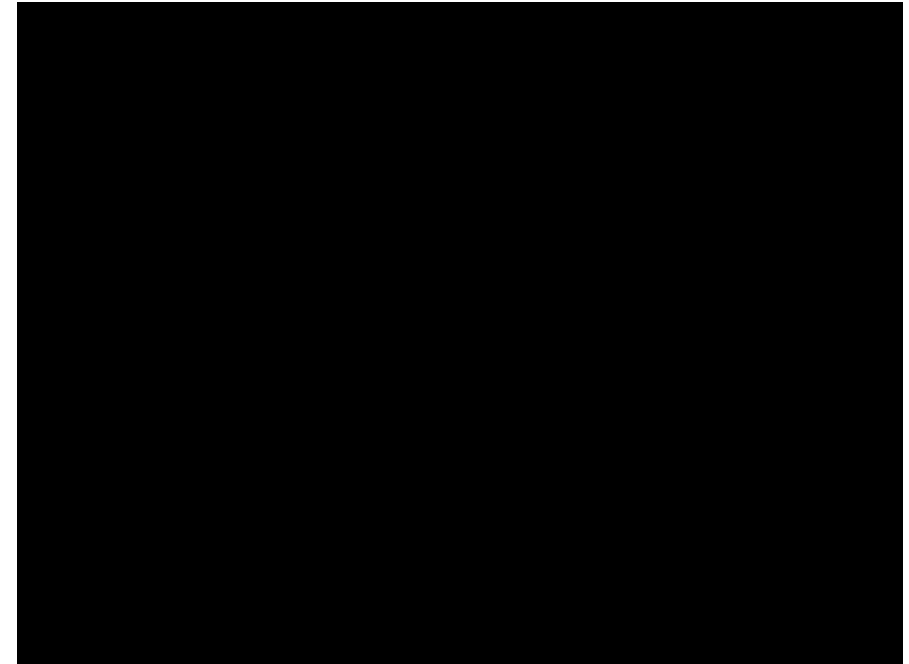
- Bearing (1)
- Seal (2)
- Reservoir (3)



RC Bit Components



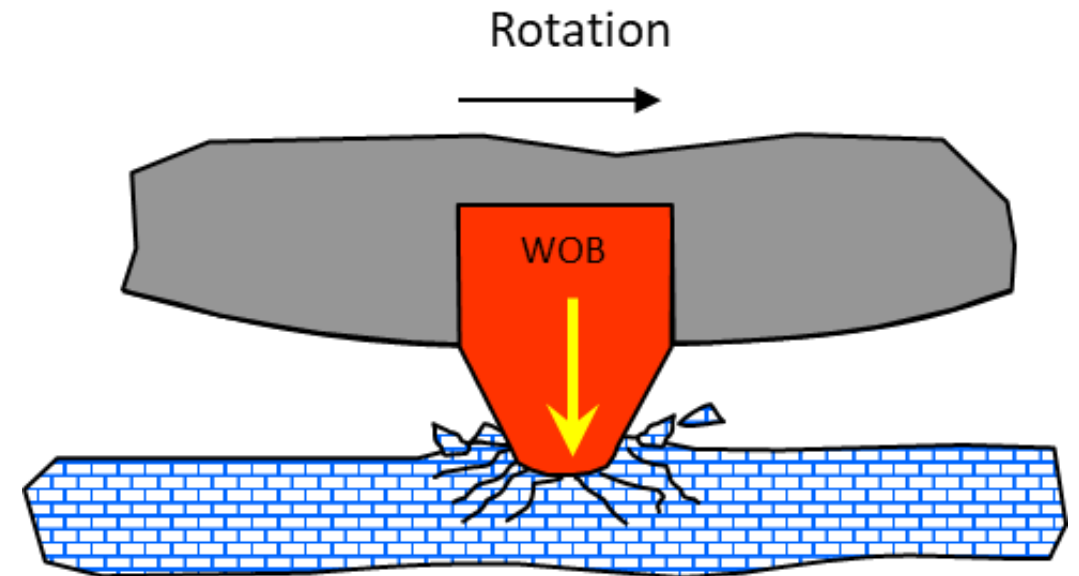
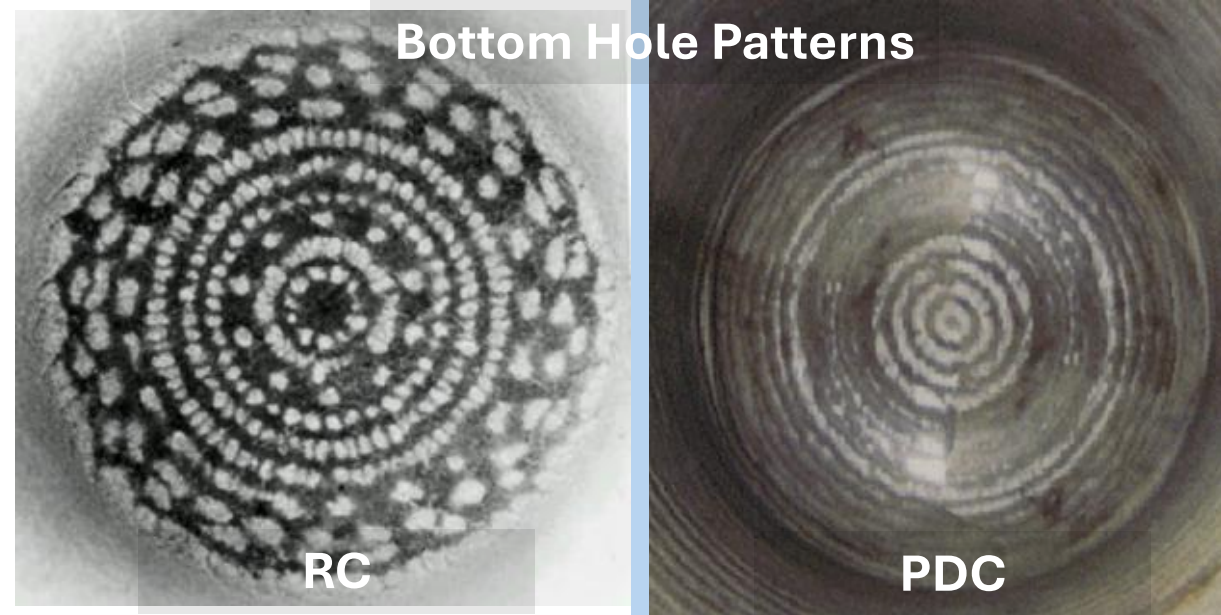
Rock Failure Type of Roller Cone Bits Or Tricone Bits



Rock Failure Mechanism

Cyclic Compression

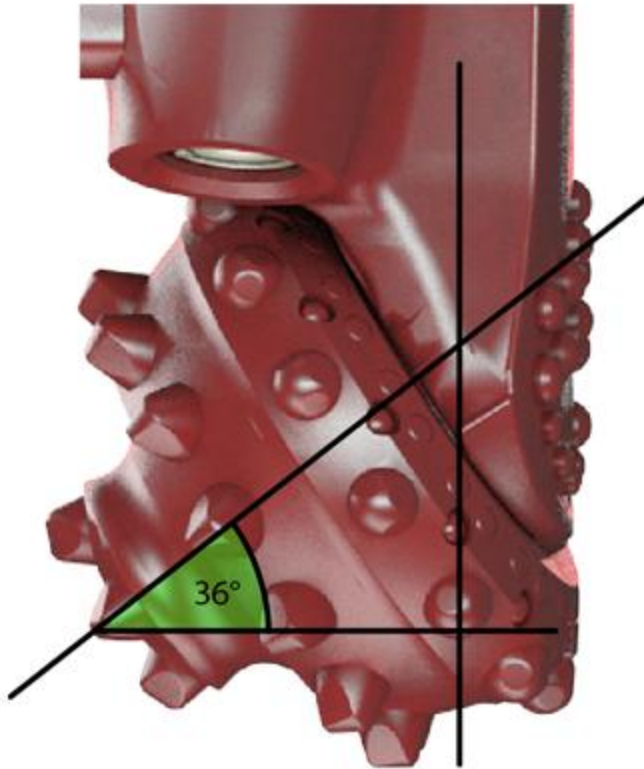
- Tooth load overcomes rock compressive strength to generate a crater – generates cracks from impact point
- Scraping helps to remove chips from craters
- Removal of chips is required to allow for new chip formation
- Hydraulics help to remove these cuttings



RC Design: Journal Angle

36 Degrees

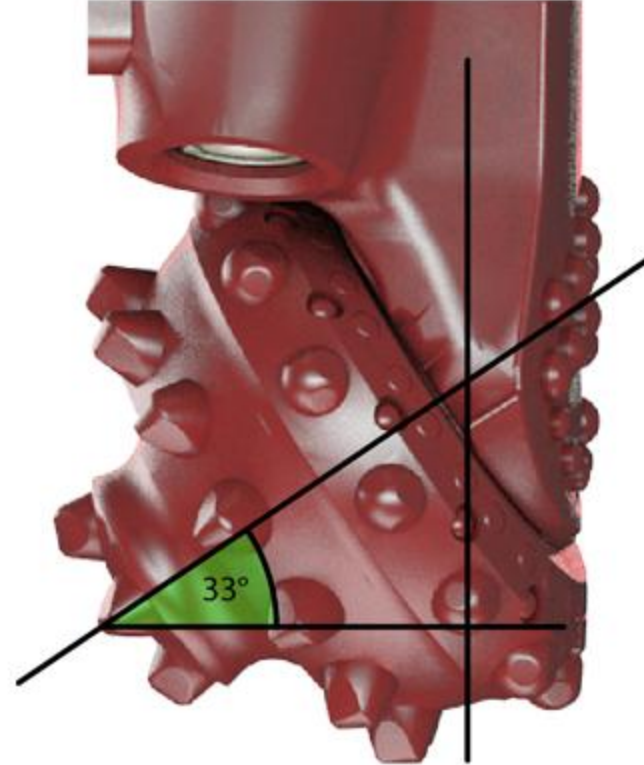
Harder Formations



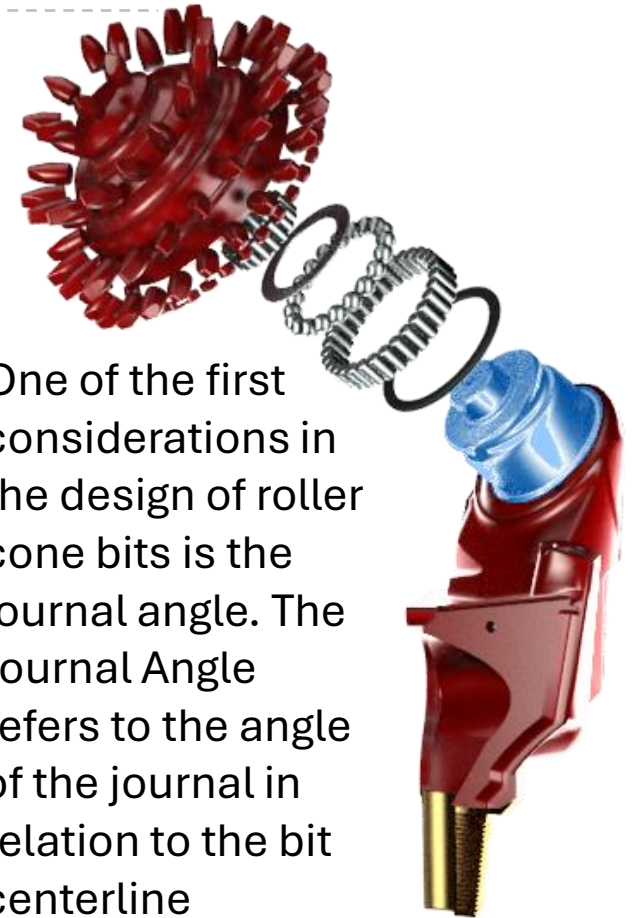
The larger journal angle allows a greater thrust load to be applied to the bit, which is desirable in hard formations. It allows for profile changes to support gauge protection.

33 Degrees

Softer Formations



The smaller journal angle provides more space for the cutting structure, which is a benefit for soft formation bits, as this allows for more of the cutting structure to engage the formation, increasing the rate of penetration.



One of the first considerations in the design of roller cone bits is the journal angle. The Journal Angle refers to the angle of the journal in relation to the bit centerline

cutters.



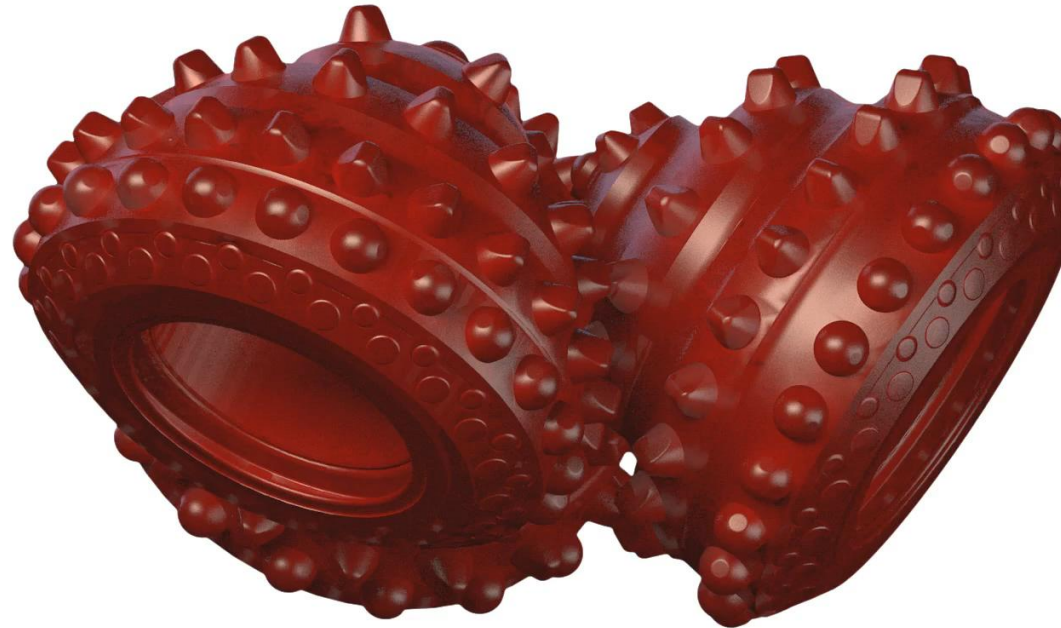
3°

5°

More aggressive cutting motion by increasing the sliding/scraping action

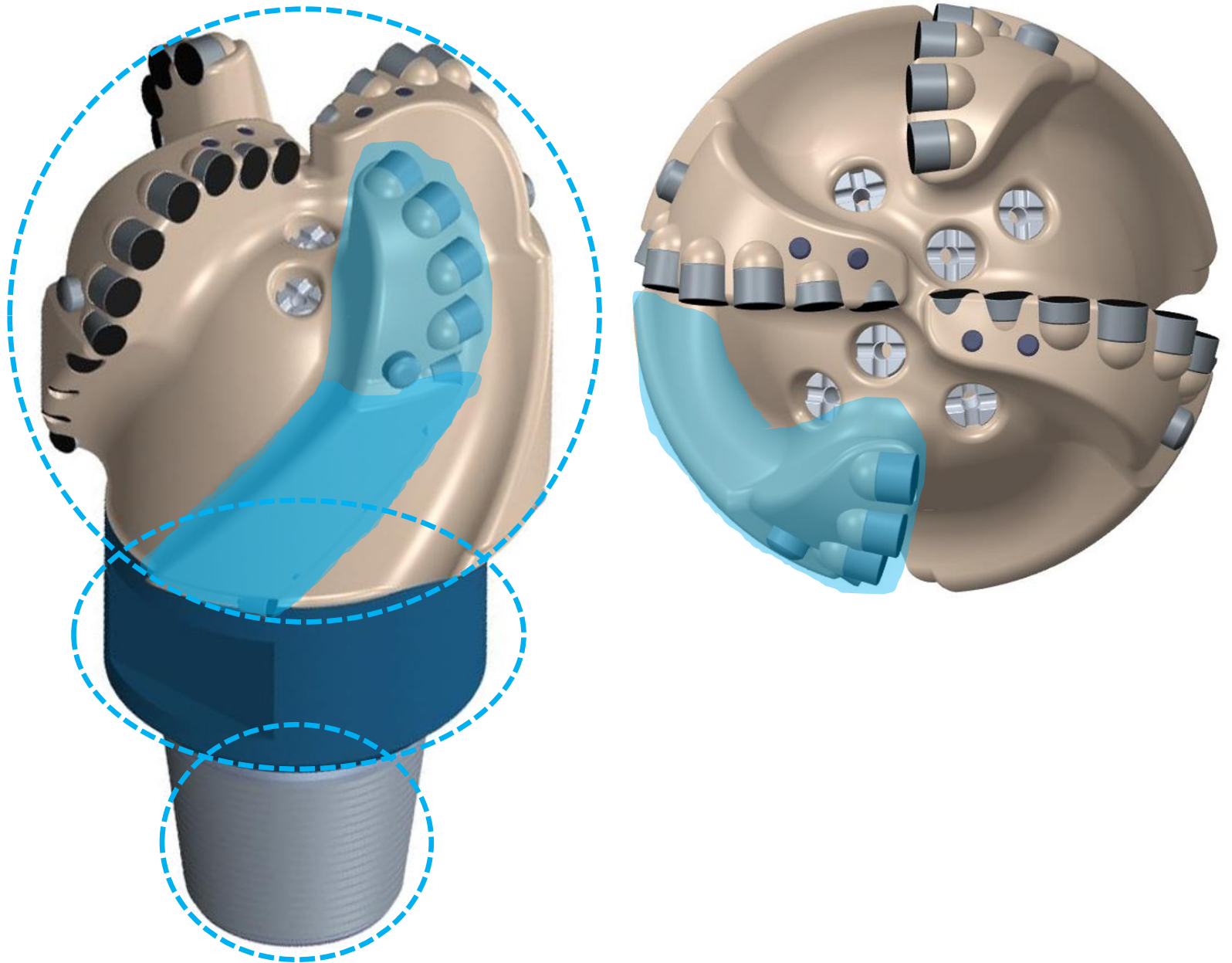
RC Design: Insert Layout

The arrangement of the cutters are laid out to ensure sufficient bottom hole coverage to fail the rock in combination with desired aggressivity / durability.



Basic Components of PDC bit

- Bit Head
- Blades
- Gauge Pad
- Cutter (PDC)
- Shank
- Bit Breaker Slot
- Connection



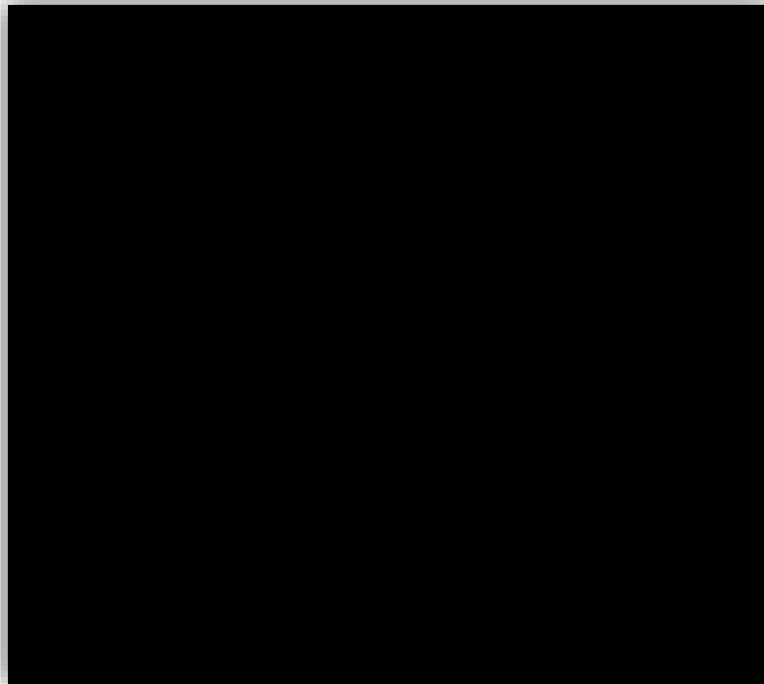
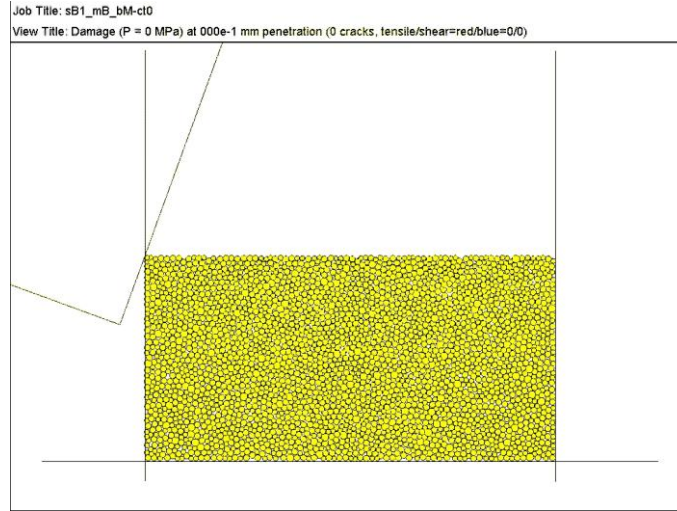
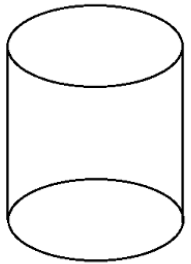
Basic Components of PDC bit



Rock Failure Mechanism

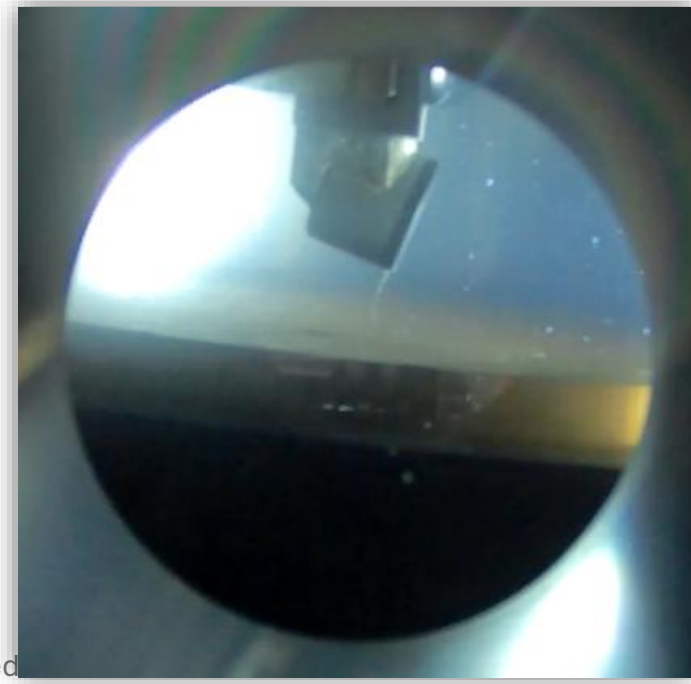
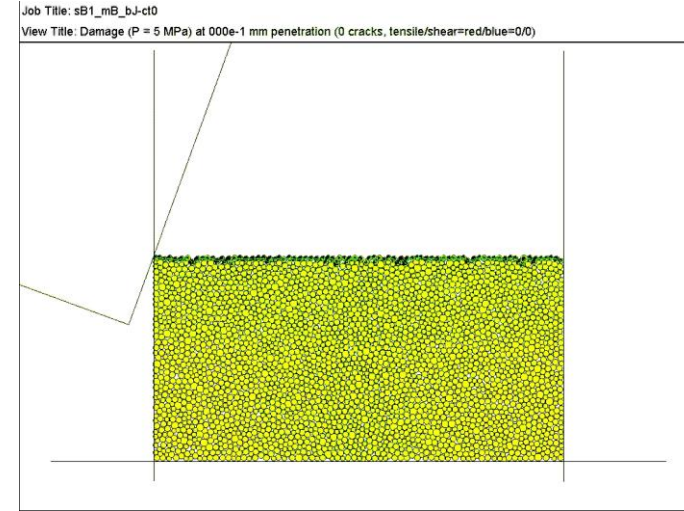
Shear

**Without
confining
pressure**

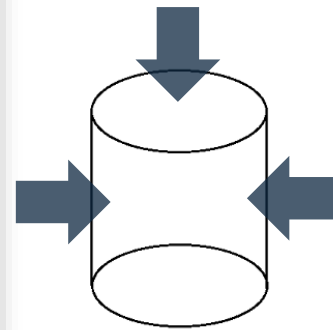


Discrete Element Method

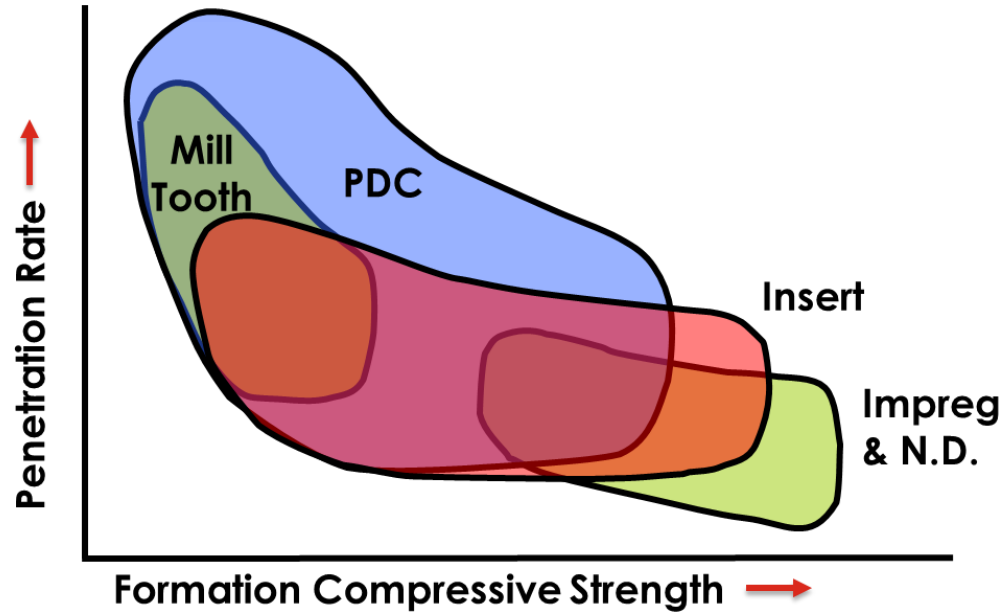
Lab Testing



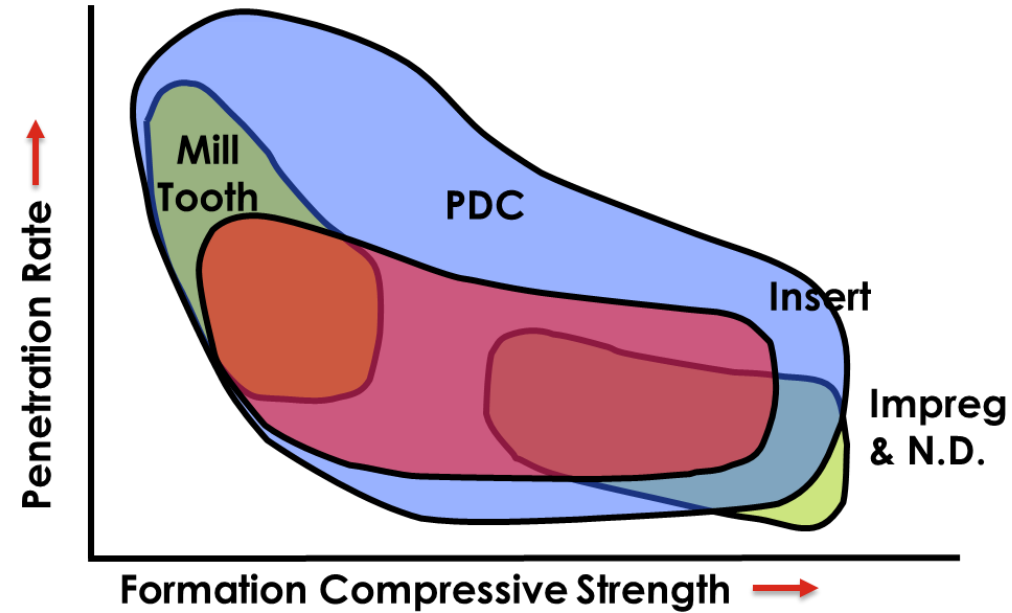
**With
confining
pressure**



Applications for Drill Bit Types



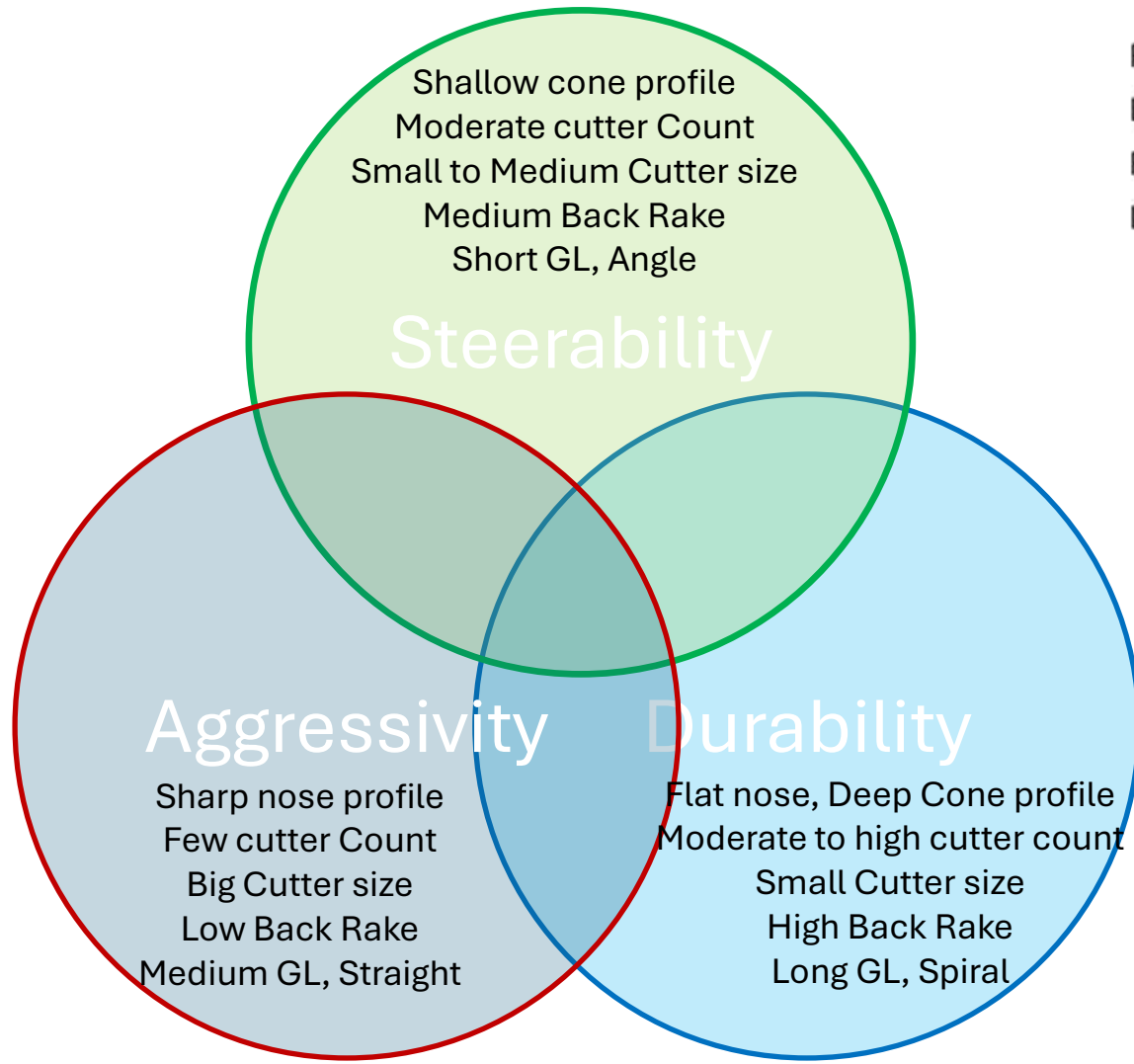
Historically



Today



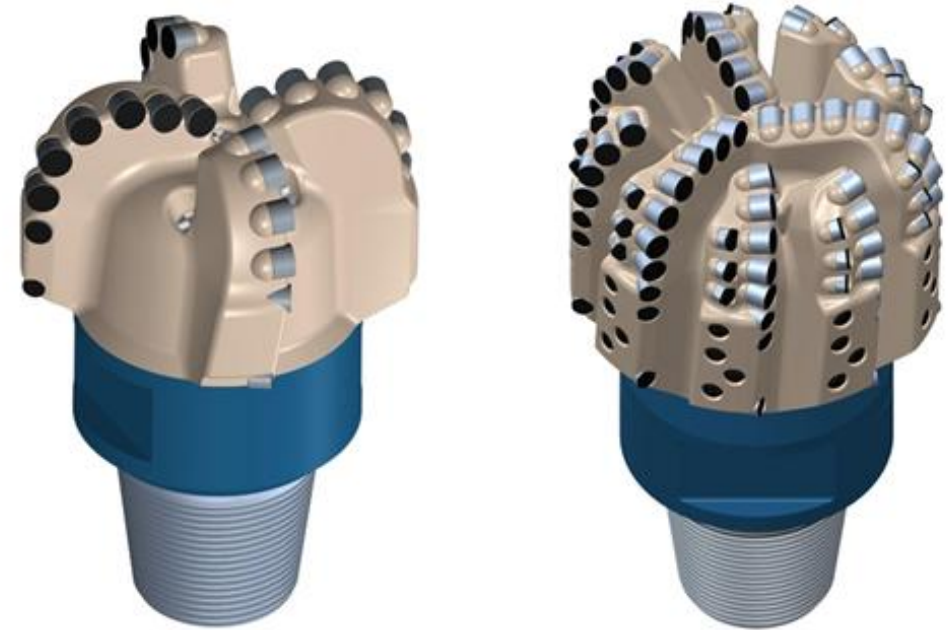
Design Compromise



FEWER CUTTERS
LARGER CUTTERS
FEWER BLADES
LOWER BACK RAKE

Aggressivity ← Durability

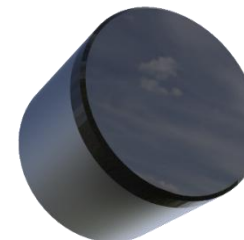
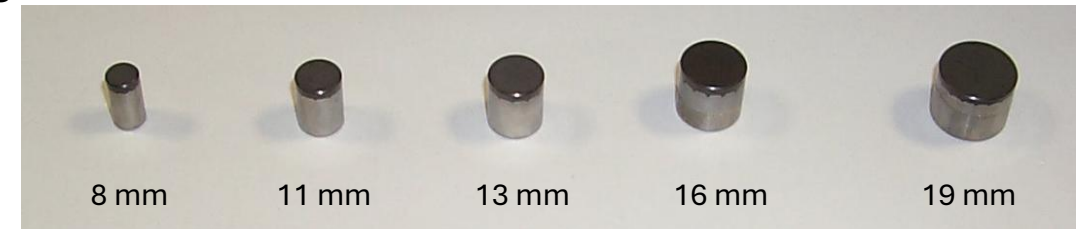
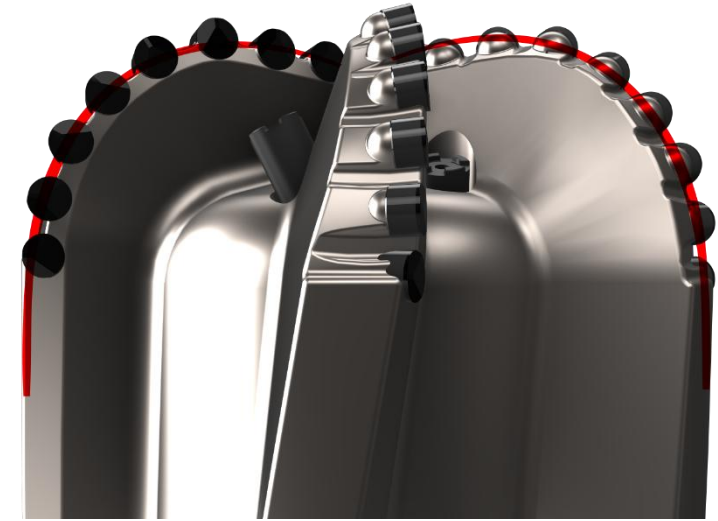
MORE CUTTERS
SMALLER CUTTERS
MORE BLADES
HIGHER BACK RAKE



Bit design compromise is between Rate of Penetration (ROP) and durability. This is because designs optimized for speed, with open and light set cutters, are often less durable, while designs for durability, with heavier, more complex cutter structures, may have a lower ROP

Blade and Cutter Configuration

- 1) The number of blades is fundamentally based on the number of cutters the design requires. The higher volume of diamond requires more blades to place the cutters along
- 2) Blade size (width) is effected by whether the design requires back-up/redundant cutting structure or secondary components
- 3) Cutter size (8mm – 25mm available)
 - 1) 13mm cutters – commonly used for smaller hole sizes or where higher impact resistance is required
 - 1) 16mm cutters – common size across all applications. Balance between achievable depth of cut and impact/thermal resistance.
 - 2) 19mm cutters – used to achieve a high depth of cut in softer applications. Able to achieve high rates of penetration but have a propensity to break when impacted at high DOC (front faced overload)

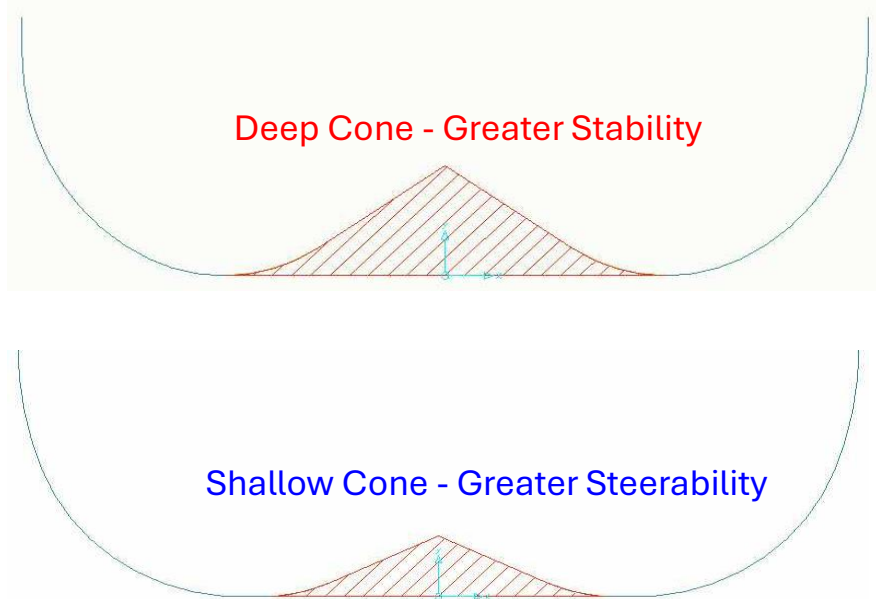


Rule of Thumb:

More Cutters = More Durability
Less Cutters = More Aggressive

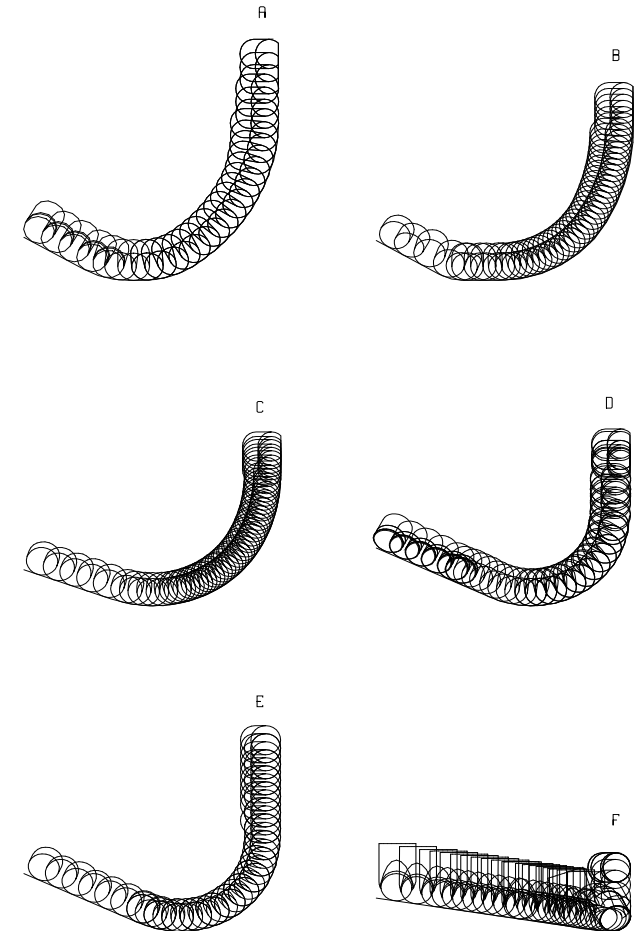
Blade Profile and Cutter Density

1) Cone Height – Deep vs Shallow effects stability and steerability



2) Tip Profile – Steerability, ROP, Stability

Profile	Description	Better For
A	Long Taper	Stability
B	Medium Taper	Stability
C	Short-Medium	Stability, ROP
D	Short Taper	Stability, ROP, Steerability
E	Short-flat Taper	Steerability
F	Flat	Steerability (Side-tracking)



3) Cutter Density – how closely the cutters are placed along the blade. Cutters placed closer together provide greater durability

4) Shoulder length

Longer shoulder = increased durability

Flatter profile = greater steerability



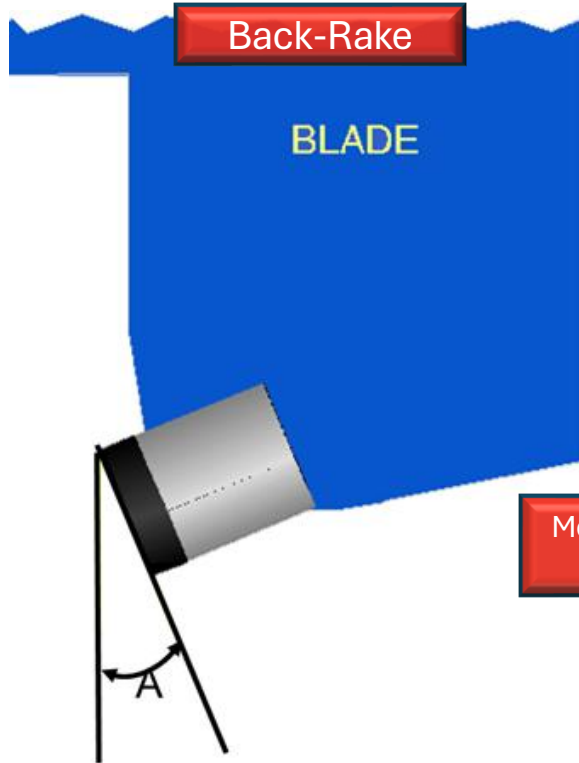
HIGH CUTTER DENSITY



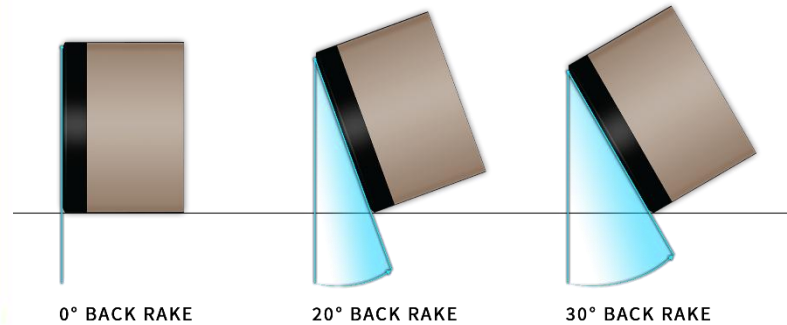
LOW CUTTER DENSITY

Cutter Rake Angles

PDC cutters are set into bits to achieve specific rake (attack) angles relative to the formation. There are two types of rake angles – back rake and side rake.

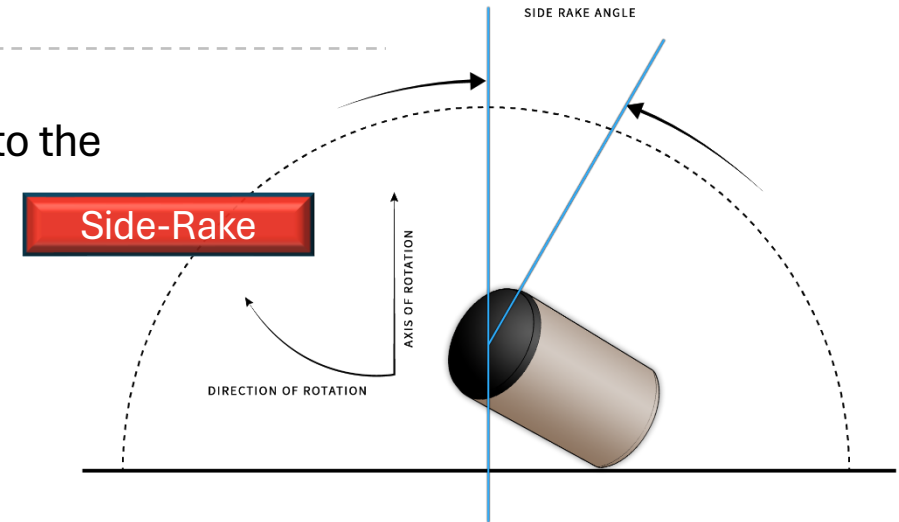
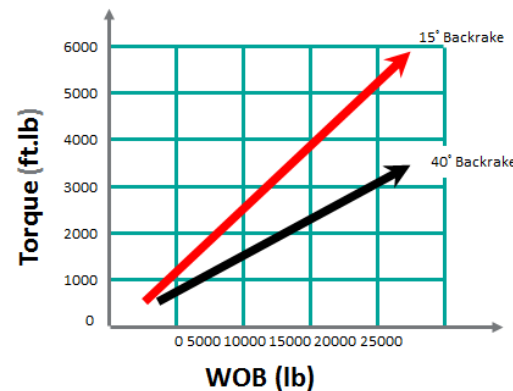


The Angle between the cutter face and a line perpendicular to the formation being drilled

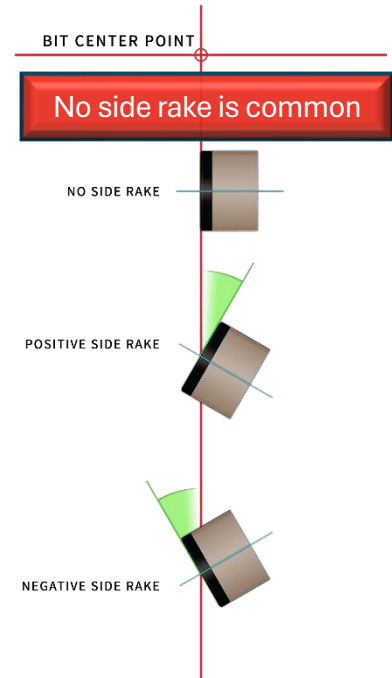


More aggressive, increase DOC, higher ROP

Less aggressive, more impact resistant, reduced DOC

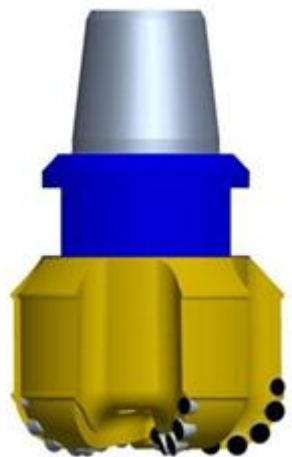


Side rake is the angle of the cutter face relative to the radius of the bit face. Negative side rake positions the cutter face towards the bit's central axes and results in cuttings being urged inward, instead of towards the junk slots, but may enhance lateral stability.



Gauge Types and Profiles

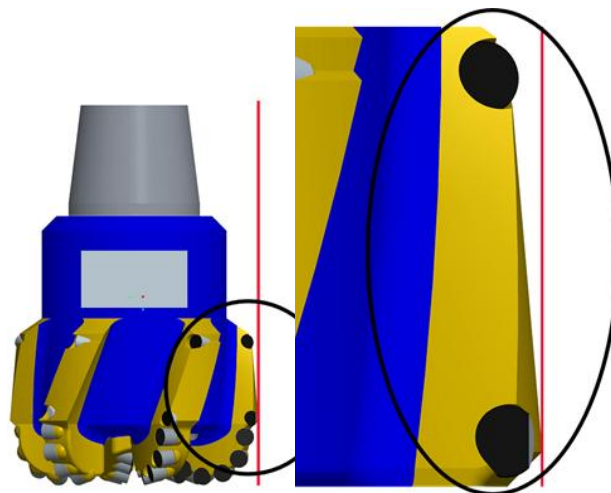
Straight Passive



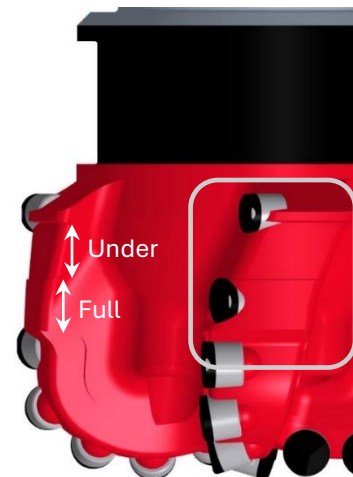
Reverse Spiral



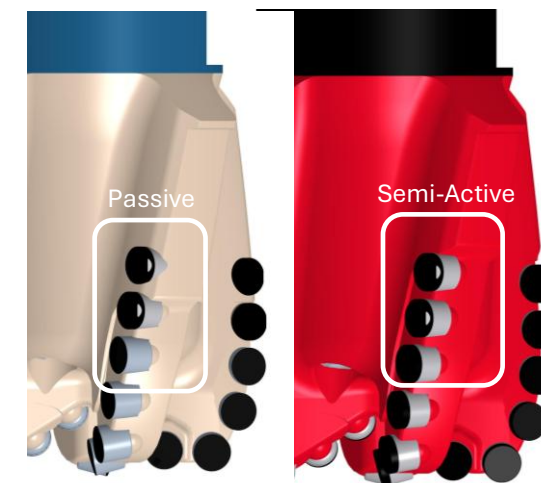
Tapered Gauge



Stepped Gauge



Semi-Active Gauge



Parameter	Straight	Reverse / Forward Spiral	Tapered Gauge
Stability	Large flat contact area however doesn't offer 360 degree contact with borehole. Can induce lateral instability and not commonly used	Enhanced borehole contact and offered continuous circumferential contact and greater stability	Commonly used for directional application due to reduced contact with the borehole wall. Little impact on stability.
Steerability	Not recommended for directional applications	Greater coverage with shorter gauge pad length improving directional control	Reduces drag from tilt by matching the angle between the wellbore and motor. Helps reduce torque and provides better directional control

Design Compromise

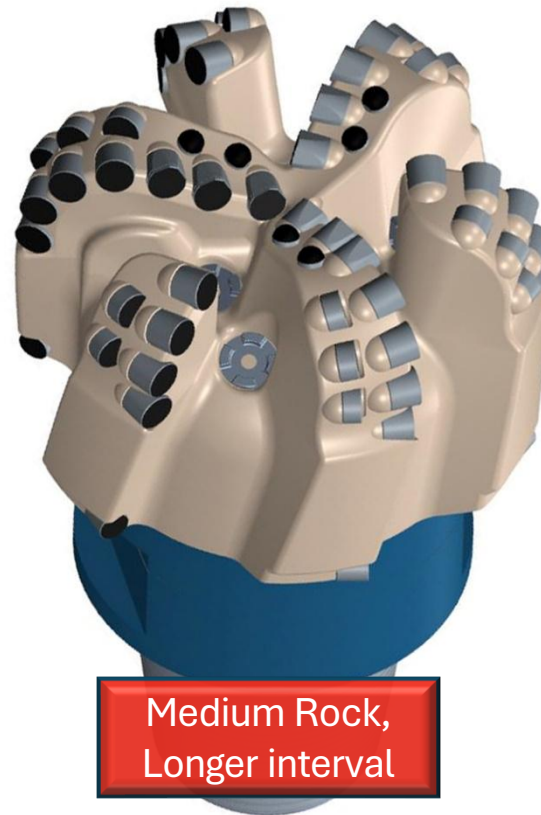
Fewer Cutters
Larger Cutters
Fewer Blades



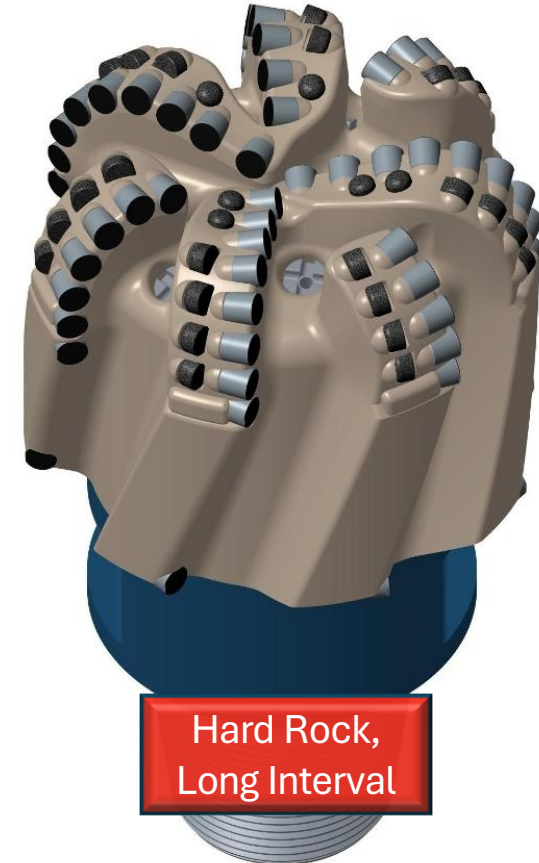
More Cutters
Smaller Cutters
More Blades



5 blade
16 mm

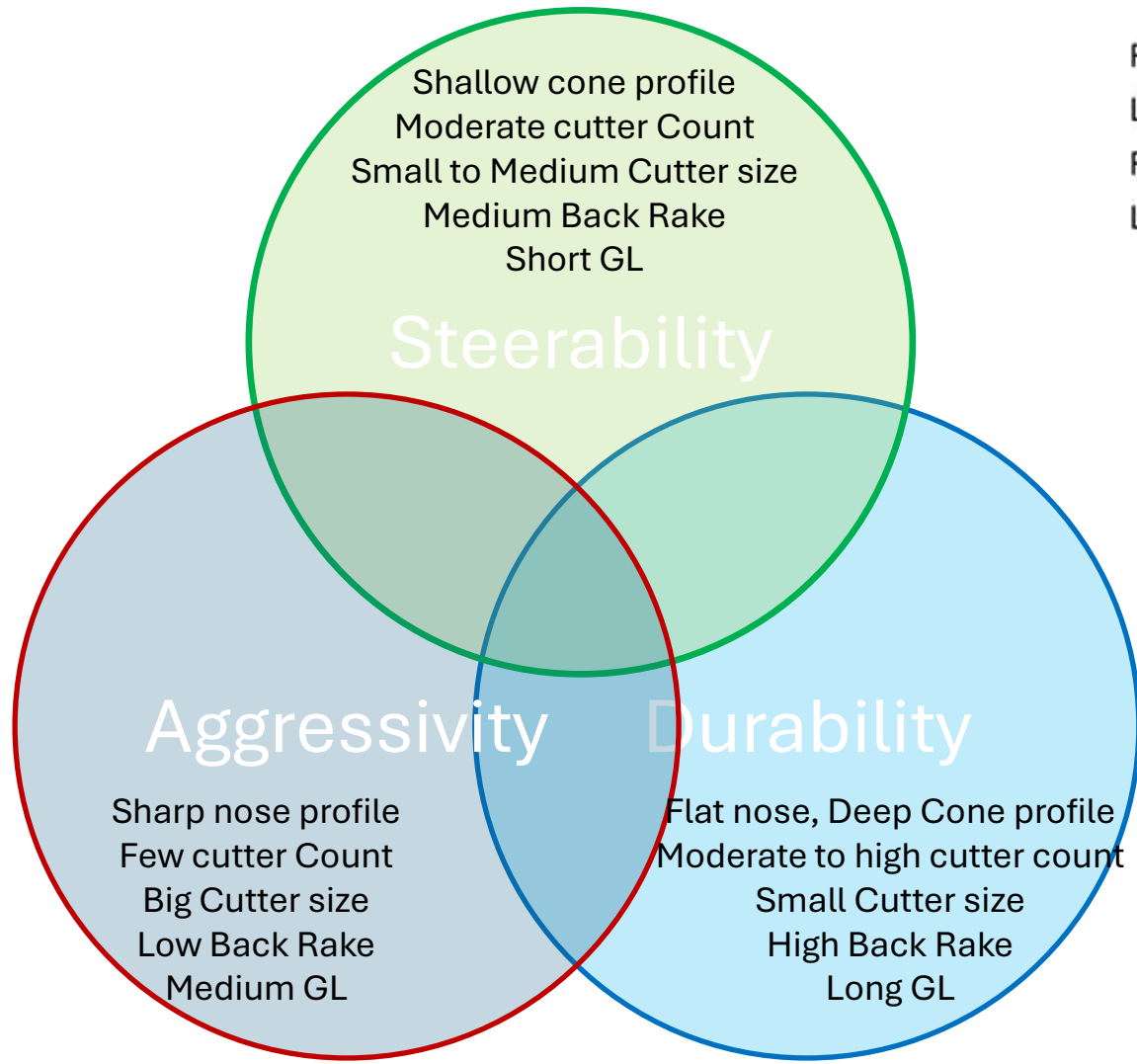


6 blade,
16 mm
double
row



7 blade 13
mm.
additional
impact
features

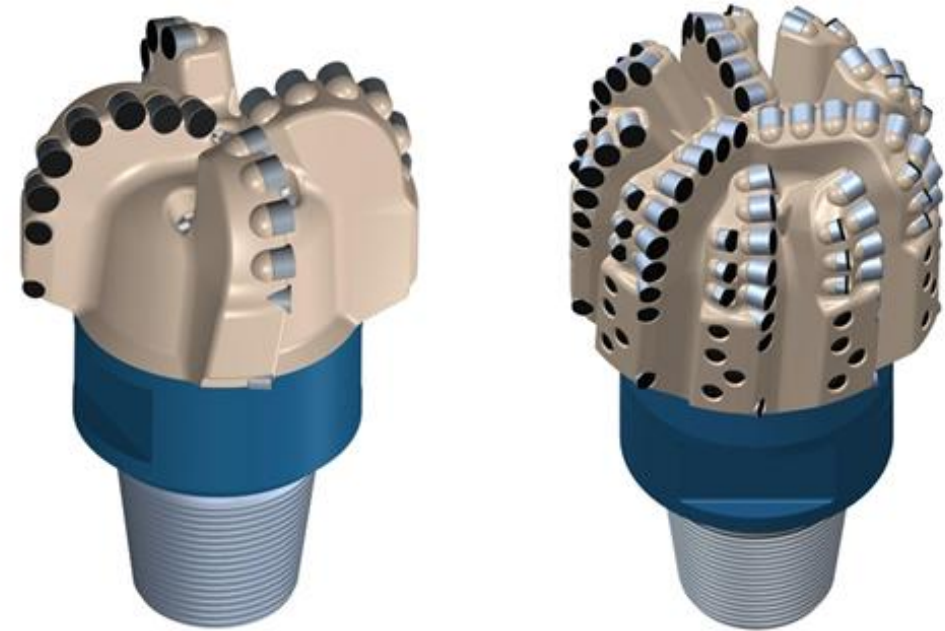
Design Compromise



FEWER CUTTERS
LARGER CUTTERS
FEWER BLADES
LOWER BACK RAKE

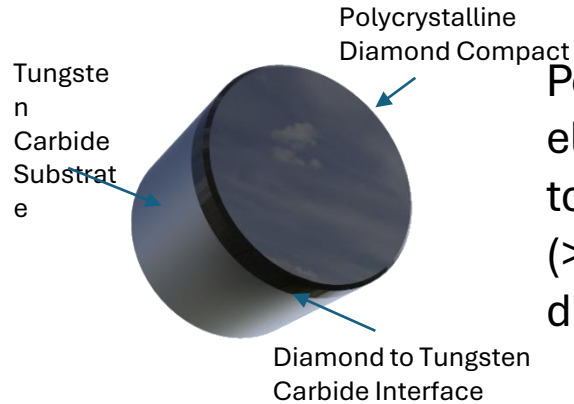


MORE CUTTERS
SMALLER CUTTERS
MORE BLADES
HIGHER BACK RAKE

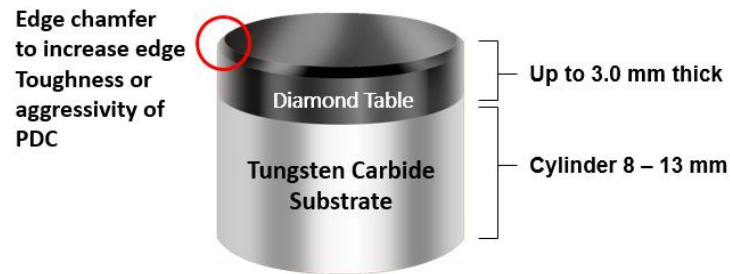
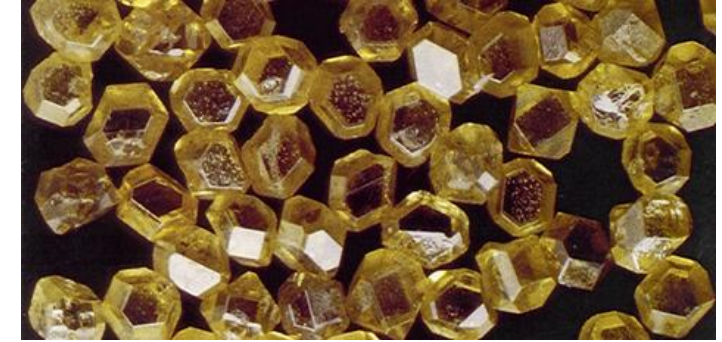


Bit design compromise is between Rate of Penetration (ROP) and durability. This is because designs optimized for speed, with open and light set cutters, are often less durable, while designs for durability, with heavier, more complex cutter structures, may have a lower ROP

PDC Cutters

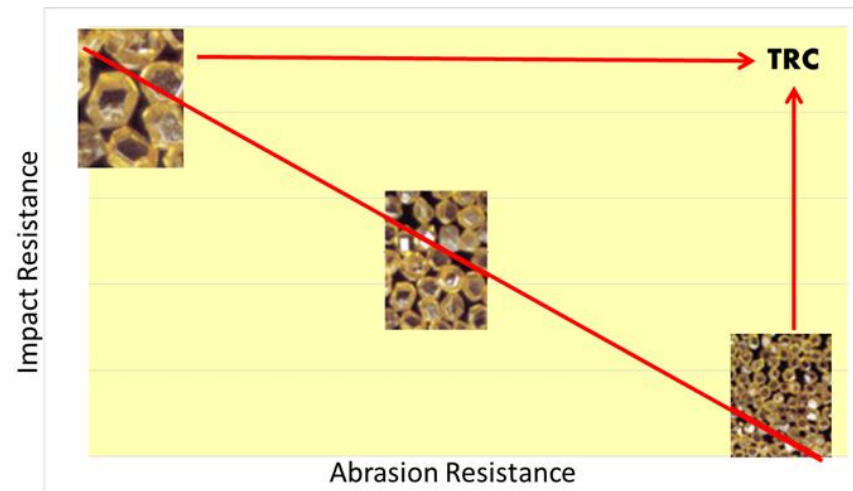


Polycrystalline diamond compacts (PDC) cutters are a cutting element composed of a polycrystalline diamond layer bonded to a tungsten carbide substrate. It is formed at high pressure ($> 1,000,000$ psi) and temperature (1300°C) from synthetic diamond powder.



PDC components consisted of a 2-3mm layer of polycrystalline diamond integrally bonded to a cemented tungsten carbide support or substrate. The face of the diamond table is then leached in hot acid to remove the Cobalt from the surface of the cutter and improve its thermal properties.

The diamond table is engineered to ensure maximum polycrystalline diamond density and to promote high abrasion and impact toughness. Larger grain = increased impact resistance
Smaller Gains = increased abrasion resistance

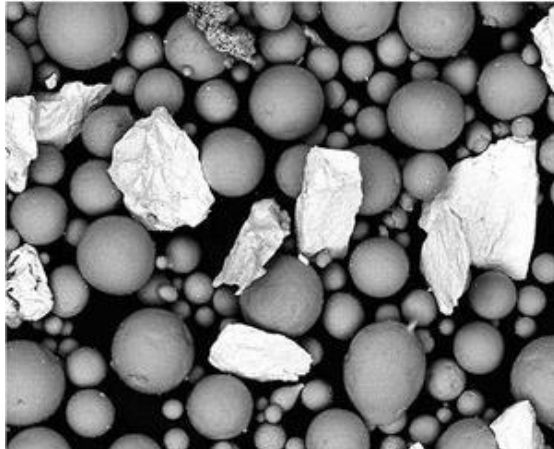


Body Material

Matrix – hard carbon molded body

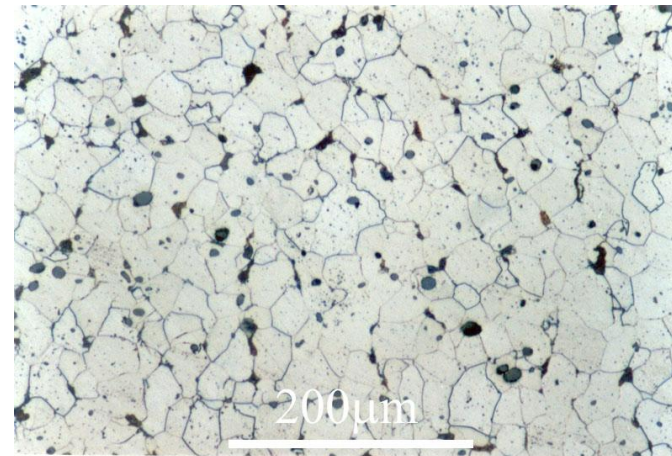


Matrix



- ✓ Shorter lead times
- ✓ Erosion resistant
- ✓ Abrasion resistant
- Limited design flexibility (blade height, cutter exposure)
- More brittle than steel
- More expensive than steel

Steel – milled body geometry



- ✓ Generally lower cost
- ✓ High shock resistance
- ✓ More aggressive due to increase cutter exposure and design flexibility
- Long lead times
- Erosion potential
- Hardfacing necessary

Break



Hydraulics and Cleaning Efficiency



Bit Cleaning / Hydraulics – OFV & JSA

Open Face Volume, (OFV) is defined as the difference in volume enclosed by the revolved tip profile and the completed bit, up to the gauge ramp.

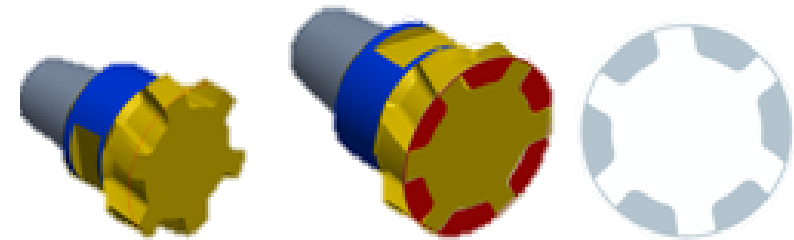
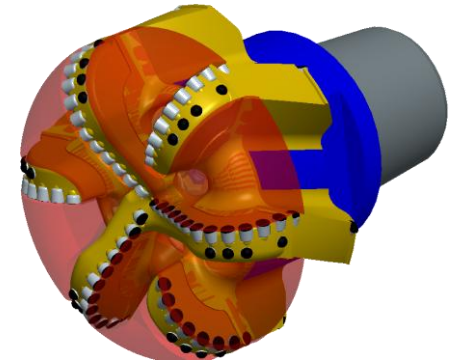
$$\text{Normalised Face Volume (NFV)} = \frac{\text{Open Face Volume (OFV)} * 100\%}{\text{Revolved Tip Profile Volume}}$$

Junk Slot Area (JSA) is defined as the minimum cross sectional area enclosed between the bit radius and milled/waterway profile.

Larger OFV and JSA become crucial for applications that induce bit balling:

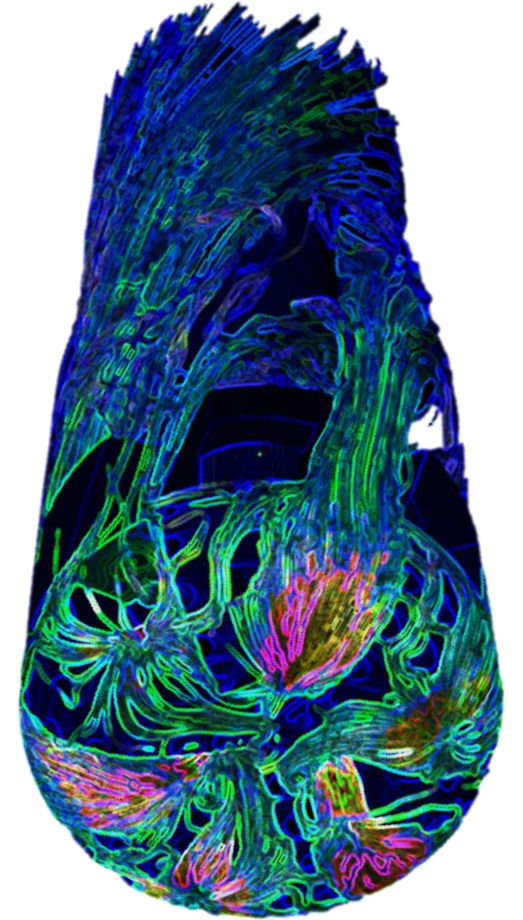
- Drilling soft, sticky formation with water based mud

- Formations react with the water in the mud, swell considerably and become sticky

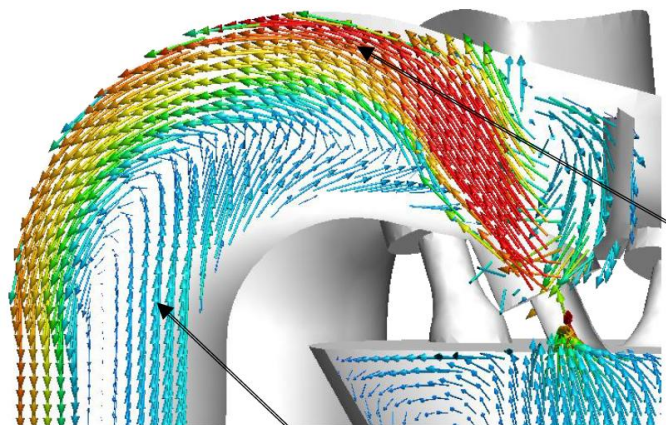


Bit Cleaning / Hydraulics - CFD

- An optimised hydraulic layout would provide cutter cooling and evacuate cuttings efficiently to avoid bit balling
- Computation Fluid Dynamics (CFD) can evaluate flow direction, junk slot flow rates, bottom hole stresses and bit body stresses to allow us to refine the hydraulic design
- The CFD model is applied with typical TFA and flow rates based on the application
 - Many iterations may be needed to achieve an optimised hydraulic setup
- There are many factors that can limit nozzle placement such as:
 - Blade location,
 - Manufacturability (clashing)
 - Blade strength

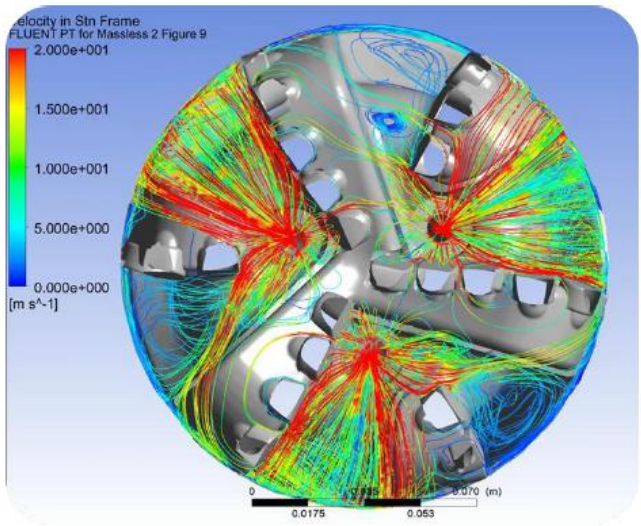


Bit Cleaning / Hydraulics - CFD



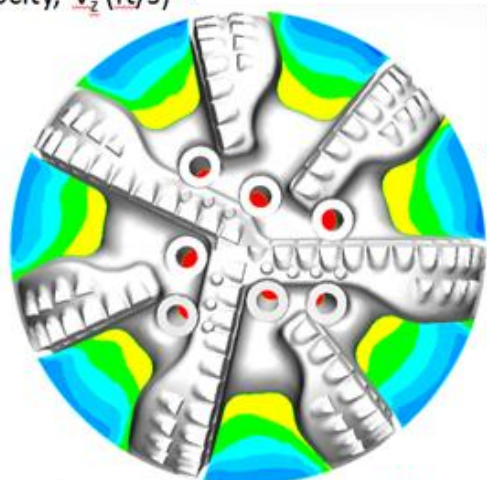
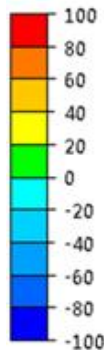
Good flow past cutters to carry cuttings off bottom hole

Recirculation zone inevitable. Smaller zone is better.

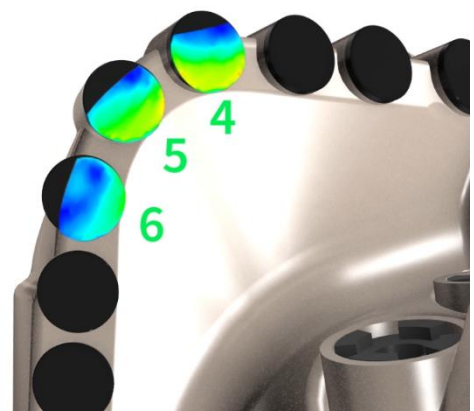


Cross sectional axial flow diagram Warmer colours are down-hole, Cooler colours are up-hole

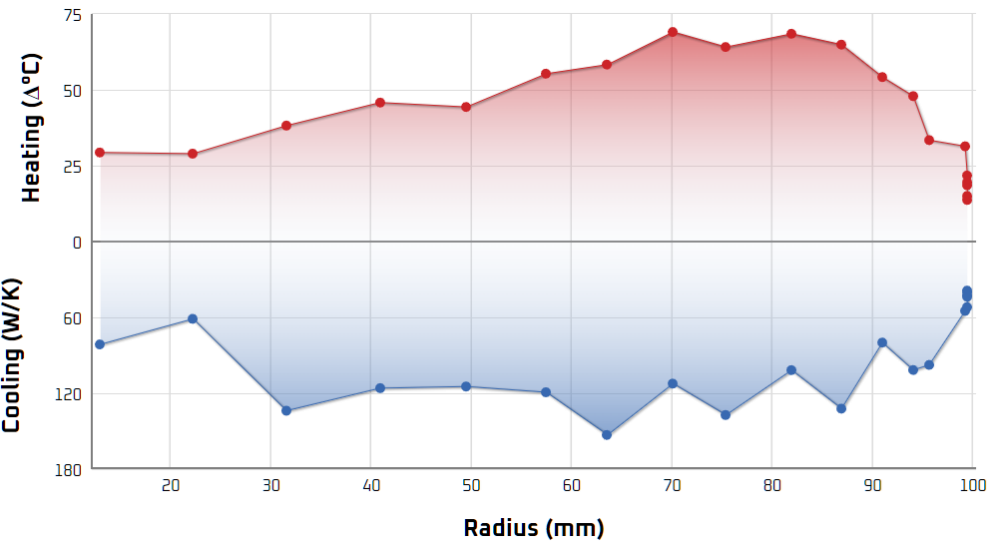
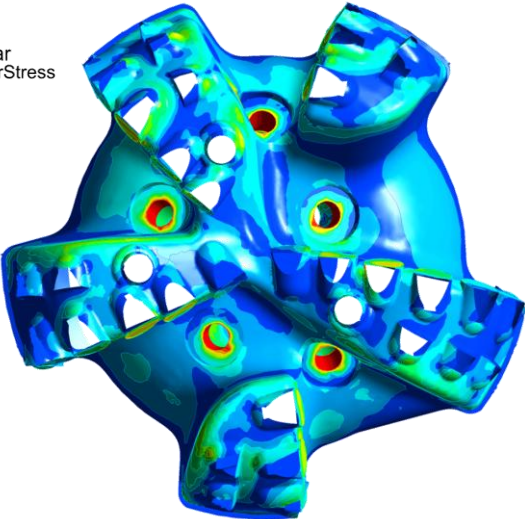
Axial velocity, V_z (ft/s)



Fluid velocities down-hole
Fluid velocities up-hole



Wall Shear
BodyShearStress
[psi]



Bit Selection General Principles



Bit Selection

The Three Principles of Bit Selection

- OFFSET DATA
- OFFSET DATA
- OFFSET DATA

OFFSET DATA

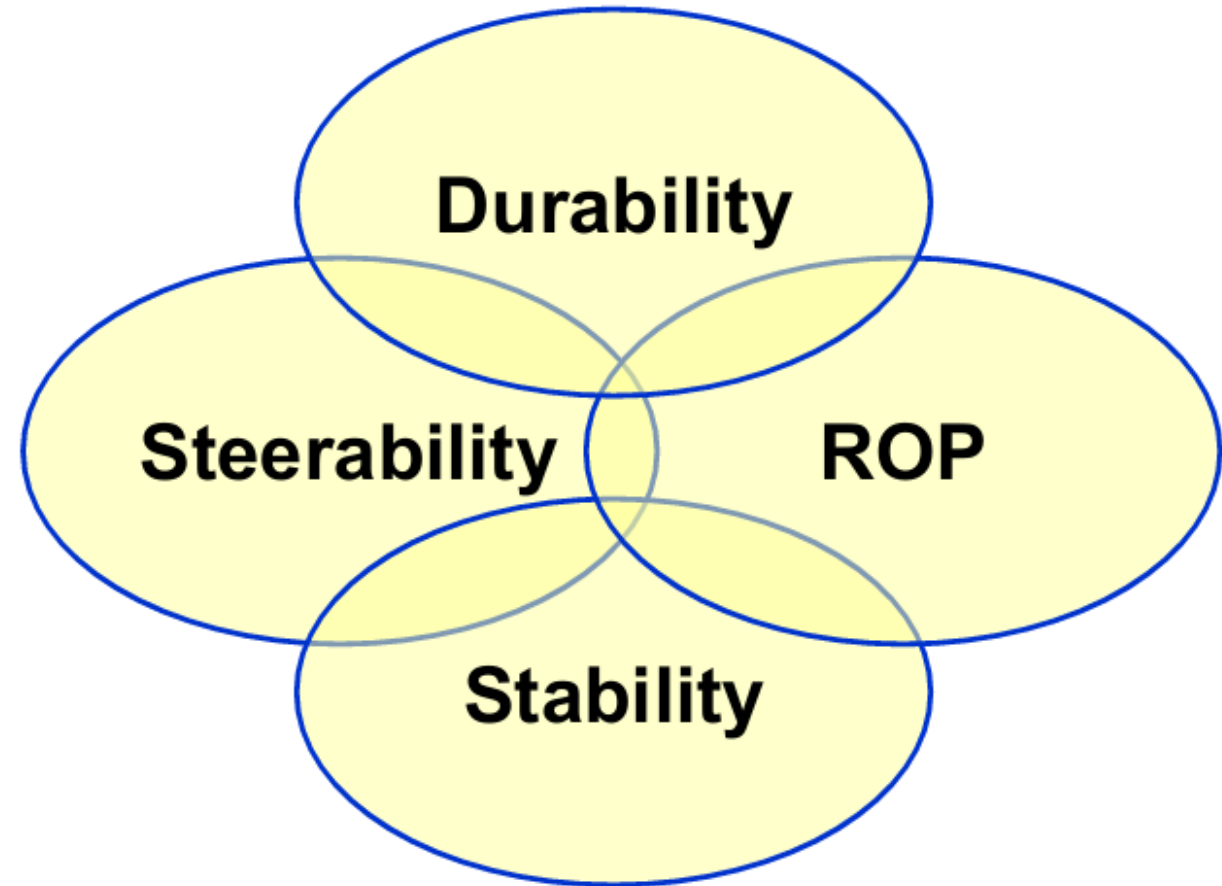
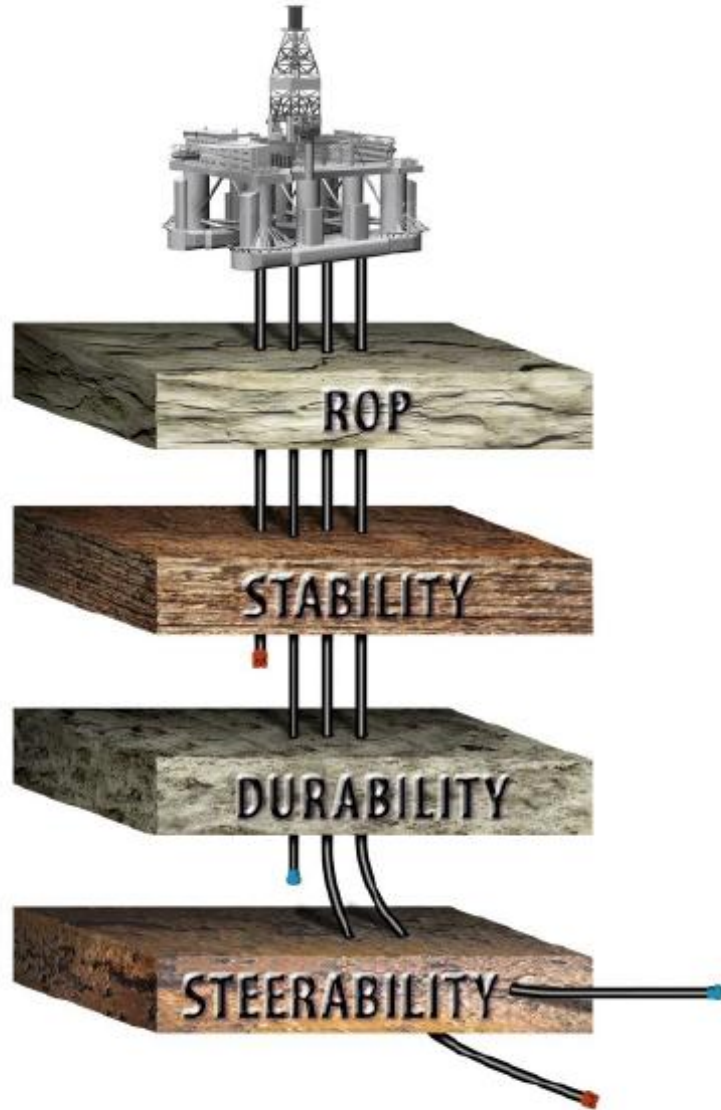
GOOD OFFSET DATA

- Bit Record
- Bits Dull Photos
- Master Logs
- DDRs
- UCS / Rock Strength

Close Offset

- Gamma Ray and Sonic
- Well Schematic or Profile
- Steering Sheets
- BHA
- Vibration Data
- End of well Reports

Drilling Target



The Application

- Lithology
- Interval
- Direction / Drive Type
- Mud
- Other

**What Do you need to know about
the application?**

Lithology

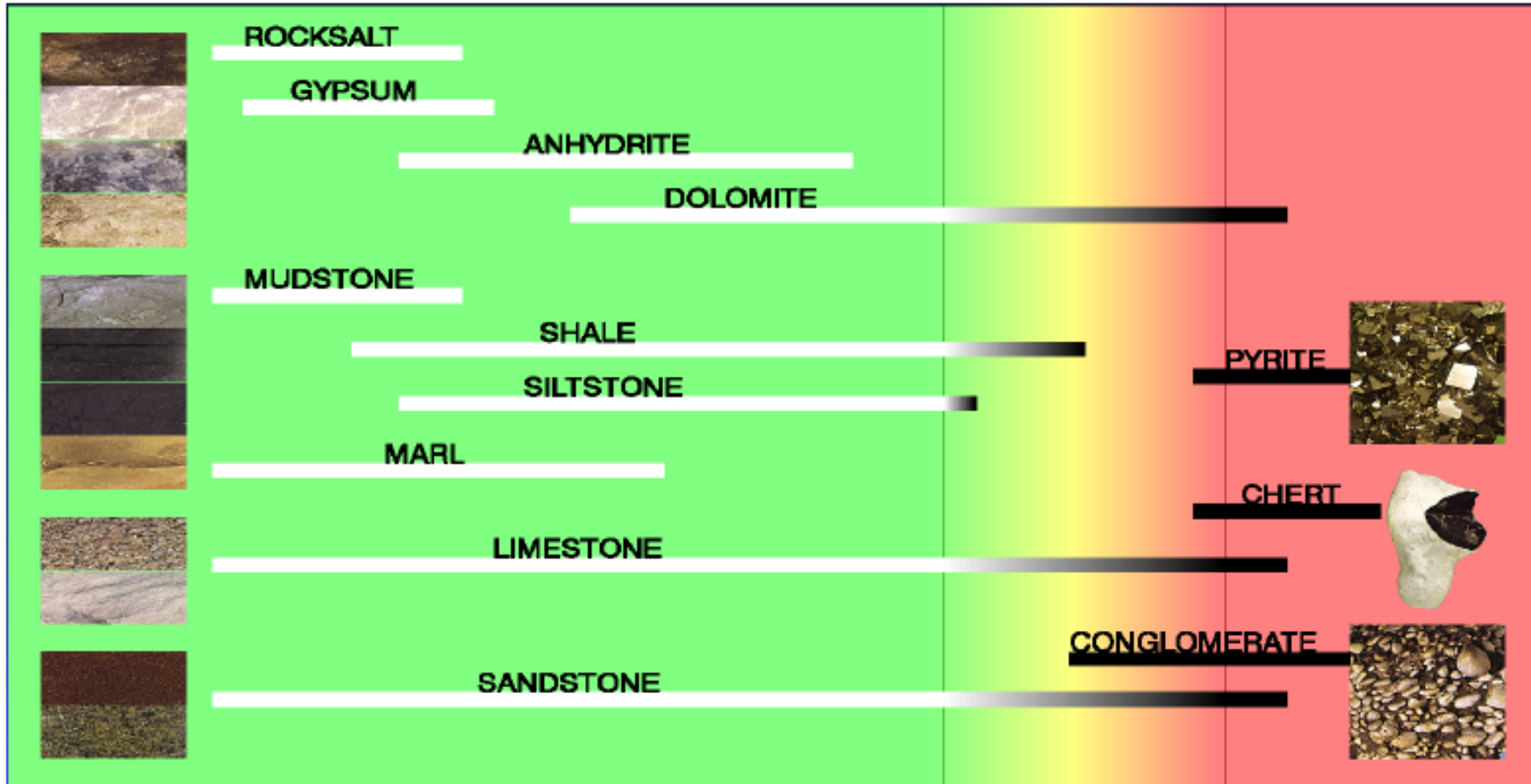
- Character
 - Rock Strength
 - Abrasiveness
 - Clay Content
 - Drillability
- Dimension
 - Thickness



Drillability

PDC Drillable

Not PDC Drillable



Pyrite



Chert



Grain Size $> 2\text{mm}$



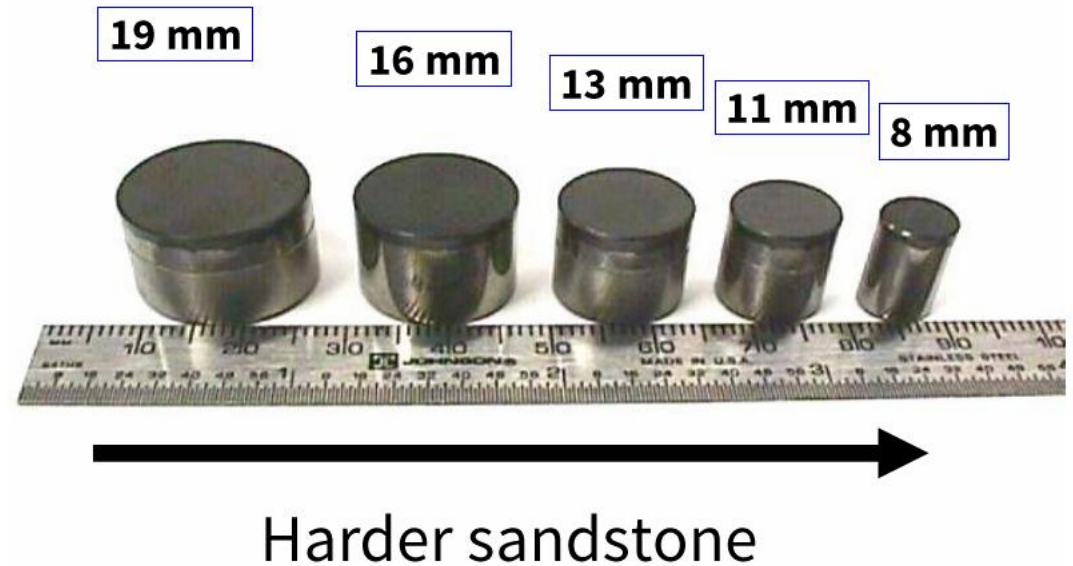
Conglomerate



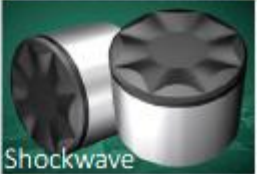
Breccia

PDC Bit in Hard Formation

- Clastic Rock: E.g. Sandstone
 - Formation is usually abrasive
- As formation becomes harder
 - Use smaller Cutter
 - More Blades
 - More Cutters



Shape Cutters technology

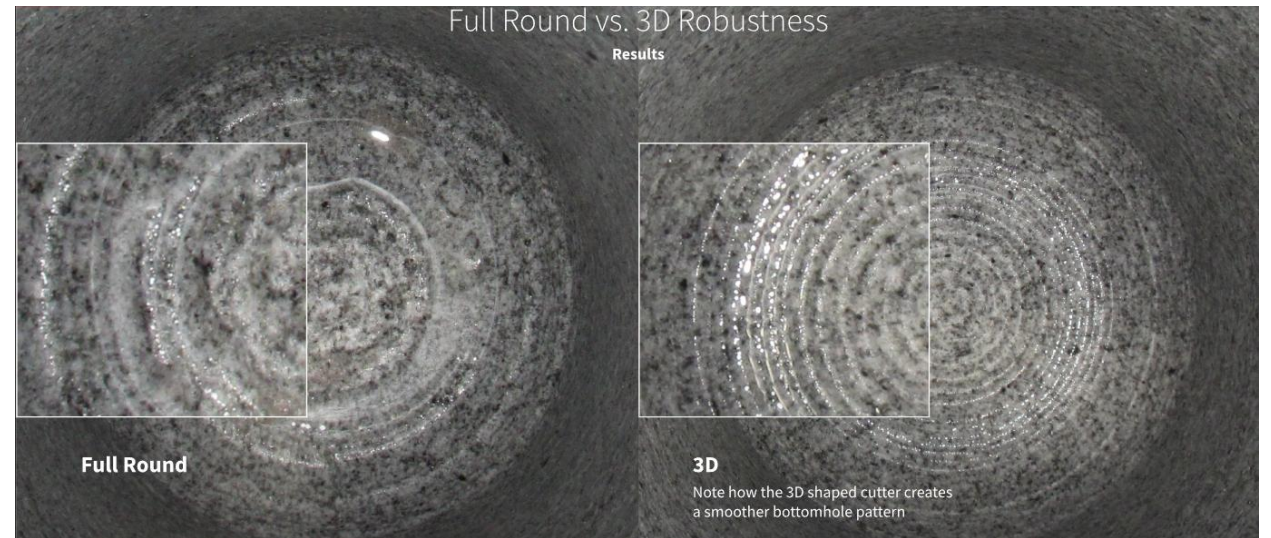
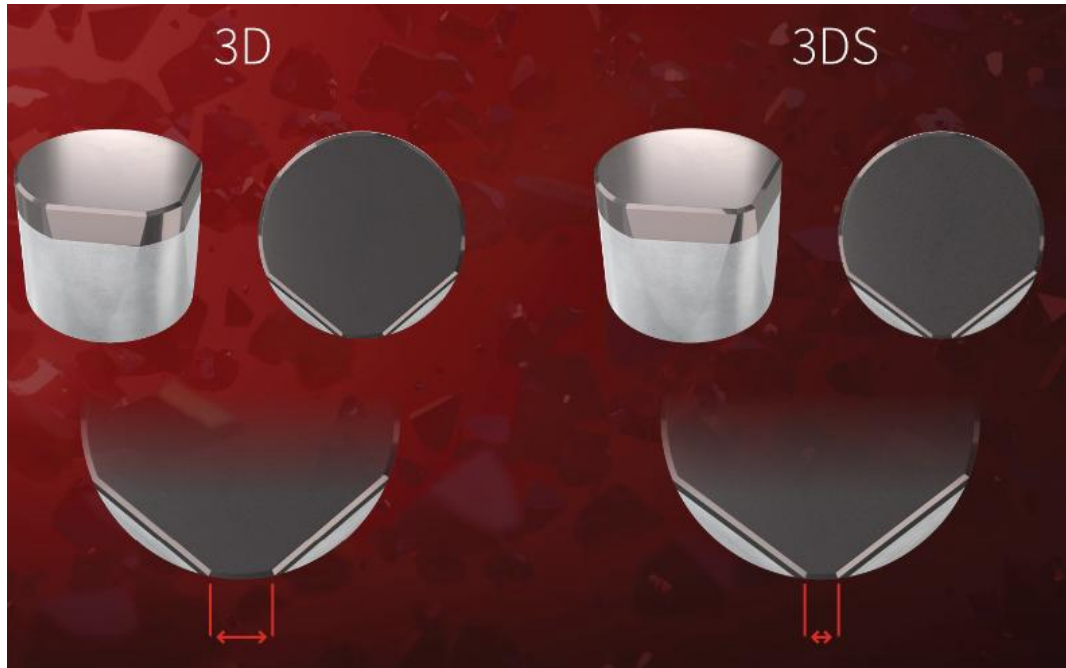
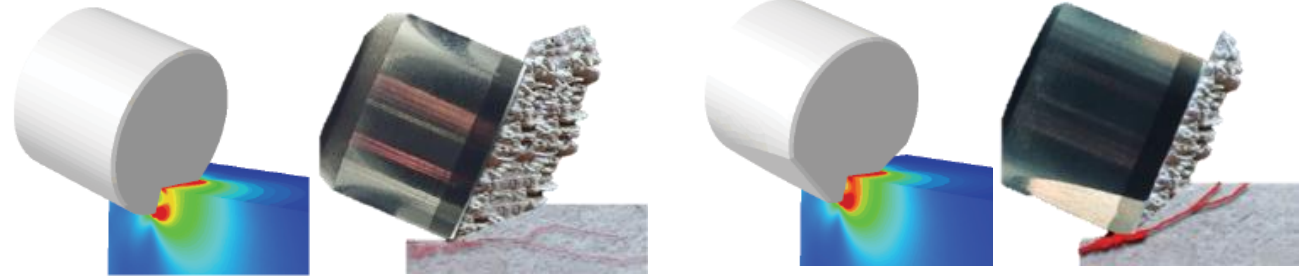
	Tough Cutters	Efficient Cutters				Point Loading Cutters		
RH	 5DX	 4DX	 ST	 STM	 STR	 3D	 4DXC	 STC
BH	 StabilisX	 Shockwave	 Staycool			 Prism	 Apex	
SLB	 Sting	 Axe Ultra	 Axe TR	 Hyperblade	 Stratablade	 Axe SR		
HAL	 Razor	 Gladius	 Dagger	 Chopper		 Machete	 Chopper SR	

Cutter Technology



ION 3D Cutter

- The chisel edge produces point loading in the WOB direction, improving DOC efficiency
- The stress increase through the stress profile and to the surface
- Helps propagate the crack to the surface, increasing efficiency in the shearing direction
- ION 3D shape – create larger, more pronounced cracks that propagate to surface requiring less energy to fail the rock while creating thinner unconsolidated cutting.

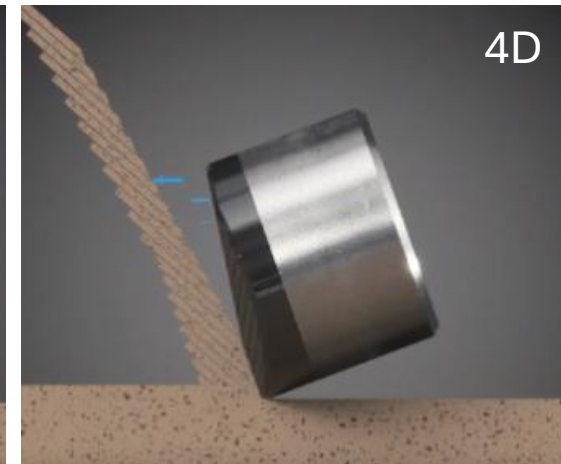


Cutter Technology



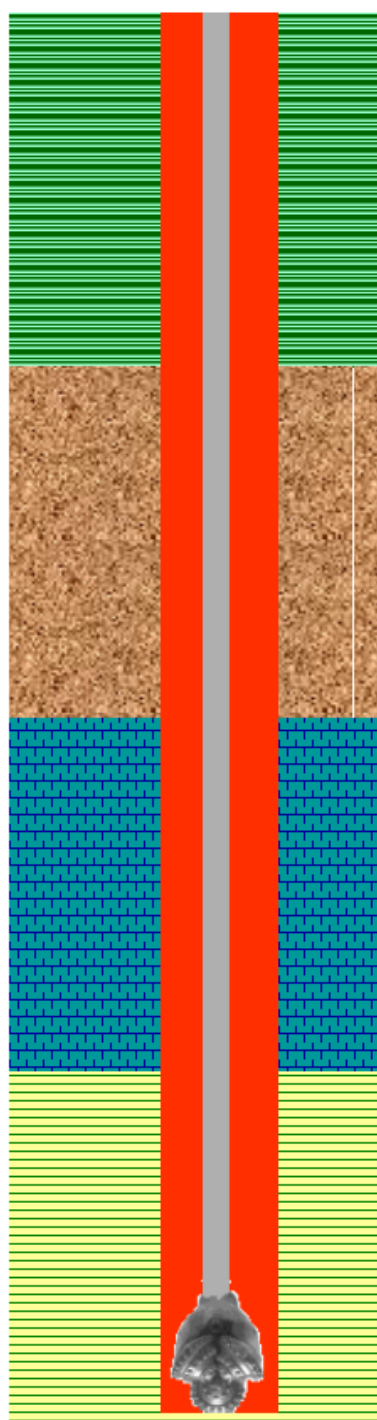
ION 4D Cutter

- 4D nonplanar cutting face create a plough effect that increase point loading in the shearing direction
- Angle rear flank create a fluid channel behind the cutting
- Decreasing adhesion to the cutter face decrease frictional losses
- Increasing hydraulic force onto the cutting breaks them into smaller pieces.

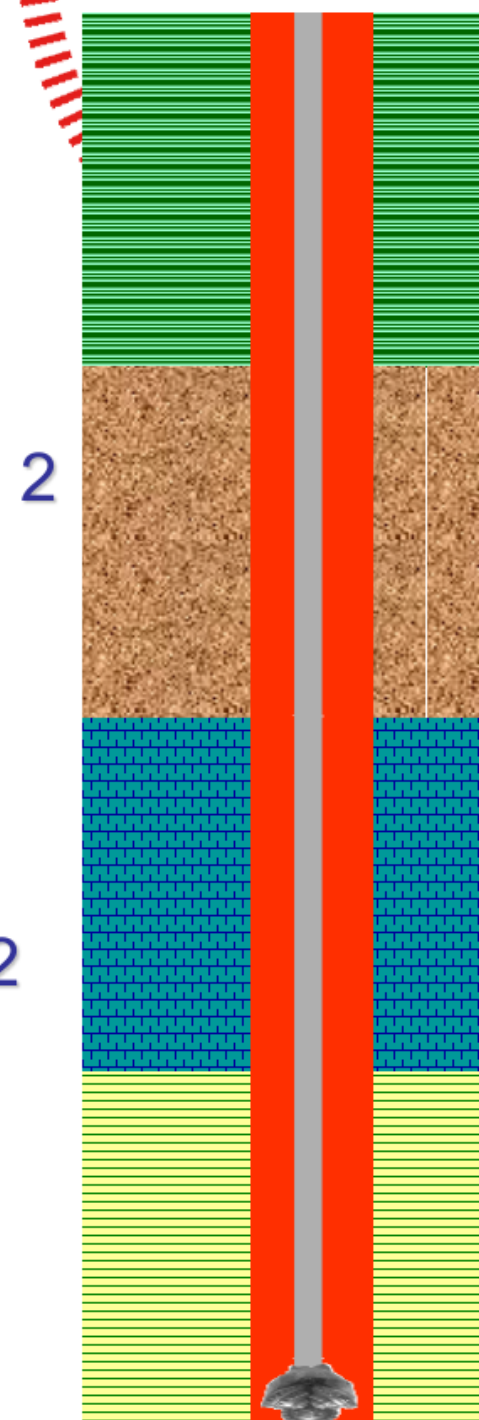


Interval

- Short or long interval?
- Single Run?
- Two Bit run?



Interval	500m	250m x 2
R.O.P.	4 m/hr	6 m/hr
Drill time	125 hrs	41.67 hrs x 2
Trip time	2 hrs	(10+2) hrs
Total Time	127 hrs	95.33 hrs
Rig cost	\$1,875 / hr	\$1,875 / hr
Bit Cost	\$40,000	\$32,000 x 2



Interval

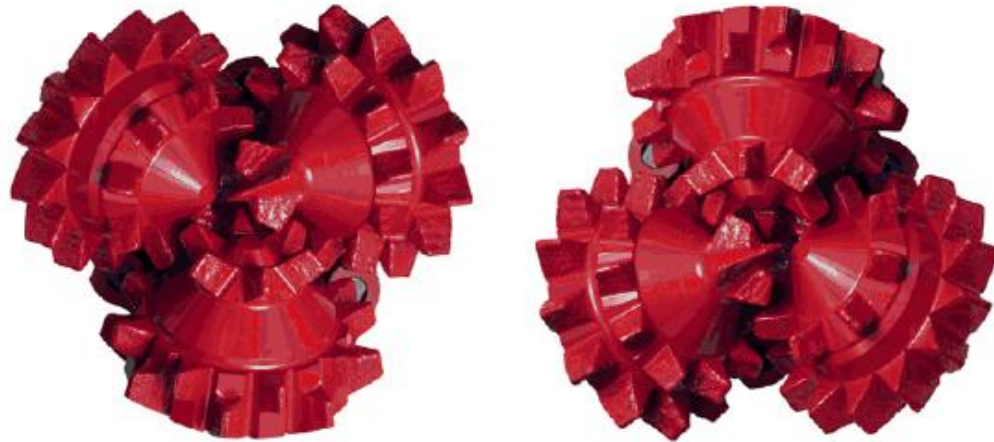
Single Run

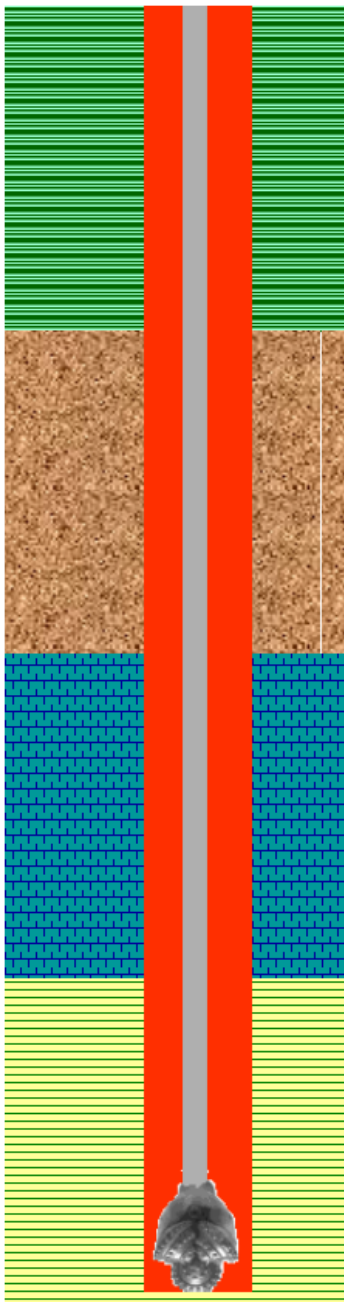
- Lower Bit Cost
- Fewer Trip



Two Bit Run

- Extra wiper trip
- Lower Total Cost



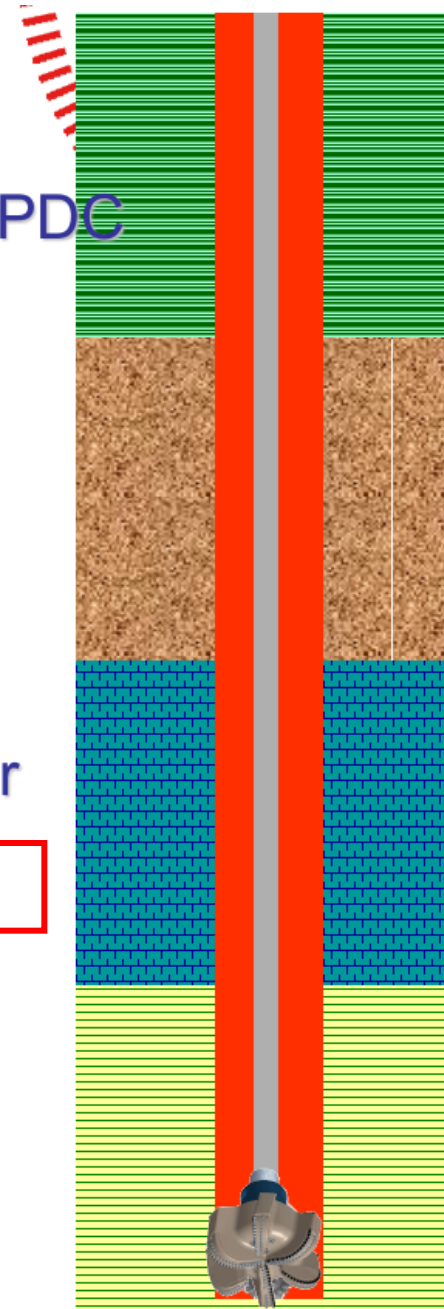


Option #1

Interval	250m x 2
R.O.P.	6 m/hr
Drill time	41.67 hrs x 2
Trip time	(10+2) hrs
Total Time	95.33 hrs
Rig cost	\$1,875 / hr
Bit Cost	\$32,000 x 2

Option #2

Interval	500m x 1 PDC
R.O.P.	9.5 m/hr
Drill time	52.63 hrs
Trip time	2 hrs
Total Time	54.63 hrs
Rig cost	\$1,875 / hr
Bit Cost	\$100,000



Interval

Single Run

- Lower Bit Cost
- Fewer Trip



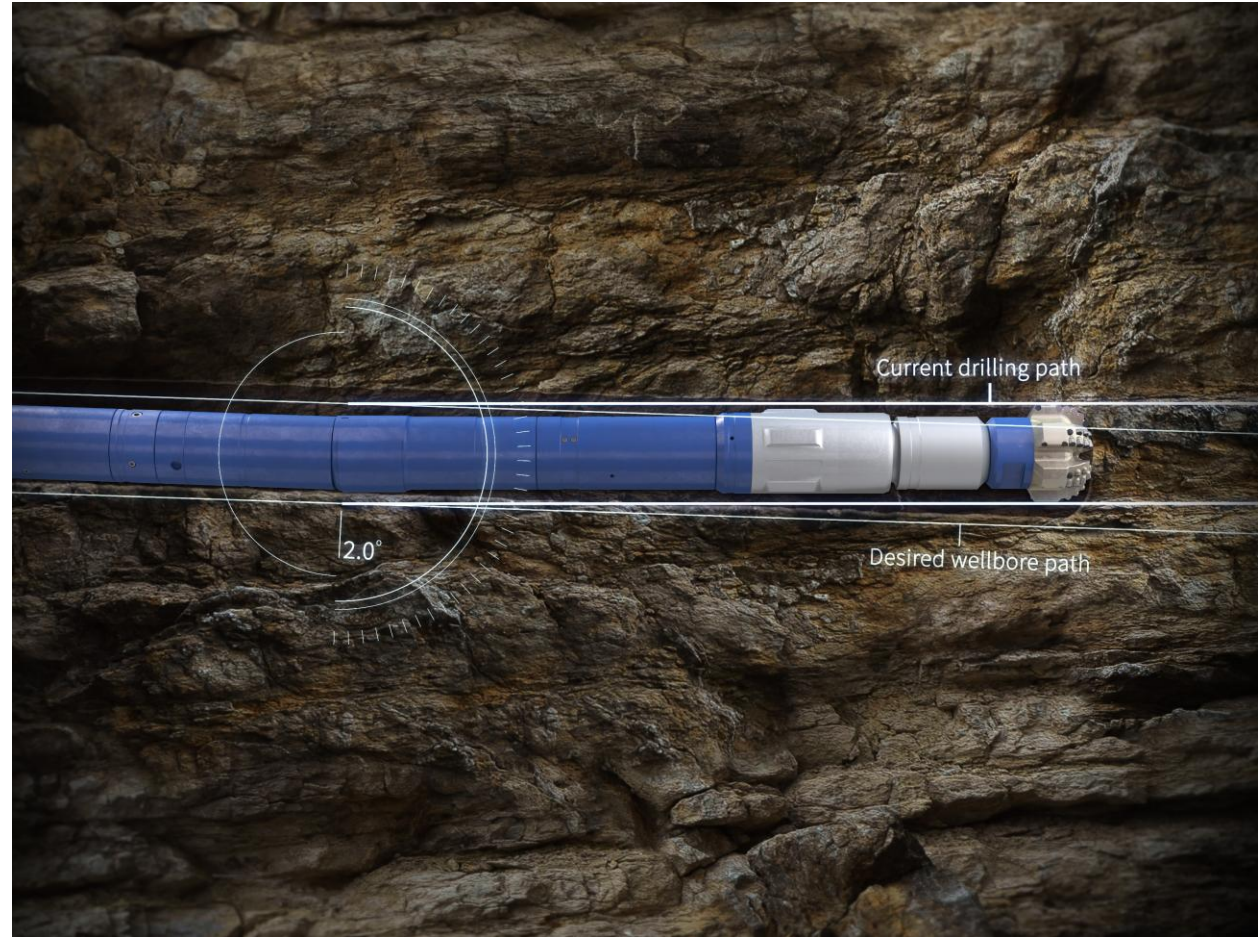
Single Run

- Higher Bit Cost But Drilling faster
- Less Drilling cost
- Fewer trip



Directional

- Well Profile
- Build/turn rates
- Drive System
 - Rotary
 - Motor
 - RSS
 - Motorized RSS



Mud Type

Water

Oil/Synthetic

**Roller
Cone**

Balling Risk

Reduced ROP

Requires - Good mud

- Good hydraulics
- 1-1 design

Reduction in seal life

Re-run issue

Reduced balling risk

Higher ROP

Longer Runs

**Fixed
Cutter**

Balling Risk

Reduced ROP

Requires – Good mud

- High JSA and OFV
- Steel body

Reduced balling risk: use
of matrix body is
practical

Other - Hydraulics

- Optimized Bit hydraulic depends of BHA used, hole depth and formation characteristic.
 - HSI: In soft formation, HSI between 3-5 is preferred for optimum ROP performance.
 - Be caution when HSI below 1 and above 5.
 - JIF: Optimized when pressure drop at the bit is around 50%
 - ✓ Large diameter shallow wells
 - ✓ Soft formations
 - ✓ Reduce bit balling in reactive formations

Other Factors

- Drill Out
 - ✓ PDC Drillable
 - ✓ Not PDC drillable
- Bi-Center Bits (pass through)
- Junk in hole (mill or RC)

Bit Selection

Summary of Selection Process

- Application
- Lithology
- Direction / Well Profile
- Interval
- Other

Performance and Failure analysis



IADC Dull Grading System

Cutting Structure							
Inner Teeth	Outer Teeth	Major Dull	Location	Seals/ Bearing	Gauge	Other Dull	Reason Pulled
☒	☒						
1	2	3	4	5	6	7	8

The SPE/IADC Fixed cutter dull grading document was last updated in 1992. Since then, the industry has adapted new, unofficial codes to overcome some the short falls in the document.

The SPE/IADC 1992 Roller cone & Fix cutter IADC dull grade documents use the same format and codes.

IADC Dull Grading for PDC

Cutting Structure						Bearing	Gauge	Remarks			
Inner Rows - Average Cutter Wear	Outer Rows – Average Cutter Wear	Primary Dull Characteristic		Location of Primary Dull Characteristic		Bearing & Seals	Undersize (16 th 's inch)	Other Dull Characteristic		Reason Pulled	
Average wear on inner 2/3 rd s of the diameter	Average wear on outer 1/3 rd of the diameter	BF	Bond Failure	C	Cone	X – Used for all fixed cutter bits	I = In-Gauge 1 = 1/16" 2 = 2/16" 3 = 3/16" ... 16 = 16/16"	BU	Balled Up	BHA	Changed Bottom Hole Assembly
		BT	Broken Cutter	N	Nose			CR	Cored	DMF	Downhole Motor Failure
		CT	Chipped Cutter	S	Shoulder			JD	Junk Damage	DSF	Drill String Failure
		ER	Erosion	T	Taper			LN	Lost Nozzle	DST	Drill Stem Test
		FC	Flat Crested Wear	G	Gauge			NR	Not Rerunnable	DTF	Downhole Tool Failure
		HC	Heat Checking					PN	Plugged Nozzle	LOG	Run Logs
		LT	Lost Cutter					RO	Ring Out	RIG	Rig Repair
		NO	No Characteristics					RR	Rerunnable	CM	Condition Mud
		RG	Rounded Gauge					WO	Wash out	CP	Core Point
		WT	Worn Cutters						See also: "Primary Dull Characteristics"	DP	Drill Plug
Value: 0-8	Value: 0-8									FM	Formation Change
0: 0% Wear	0: 0% Wear									HP	Hole Problems
1: 1/8 Wear	1: 1/8 Wear									PP	Pump Pressure
2: 2/8 Wear	2: 2/8 Wear									PR	Penetration Rate
3: 3/8 Wear	3: 3/8 Wear									TD	Total Depth
4: 4/8 Wear	4: 4/8 Wear									TQ	Torque
5: 5/8 Wear	5: 5/8 Wear									TW	Twist-Off
6: 6/8 Wear	6: 6/8 Wear									WC	Weather Conditions
7: 7/8 Wear	7: 7/8 Wear									WO	Washout Drill String
8: 100% Wear	8: 100% Wear										
4 = 4/8 = 1/2 of the cutter diameter = 50% wear	2 = 2/8 = 1/4 of the cutter diameter = 25% wear						"I" can be mistaken for "1".	Primary & Other Characteristics used for Roller Cone bits:			
							"0" and "IN" are unofficially used.	BC - Broken Cone CC - Cracked Cone CD - Cone Drag CI - Cone Interference LC - Lost Cone OC - Off Center Wear PB - Pinched Bit SD - Shirttail damage SS - Self Sharpening Wear TR - Tracking			
							Measured with a "No-Go" PDC Gauge Ring				
	*Gauge Cutters are pre-ground during the manufacturing process.		*Box is only used for cutter dull characteristic		*More than one Location can be noted						

IADC Dull Grading for Roller Cone

Inner versus Outer Teeth – Milled Tooth Bit

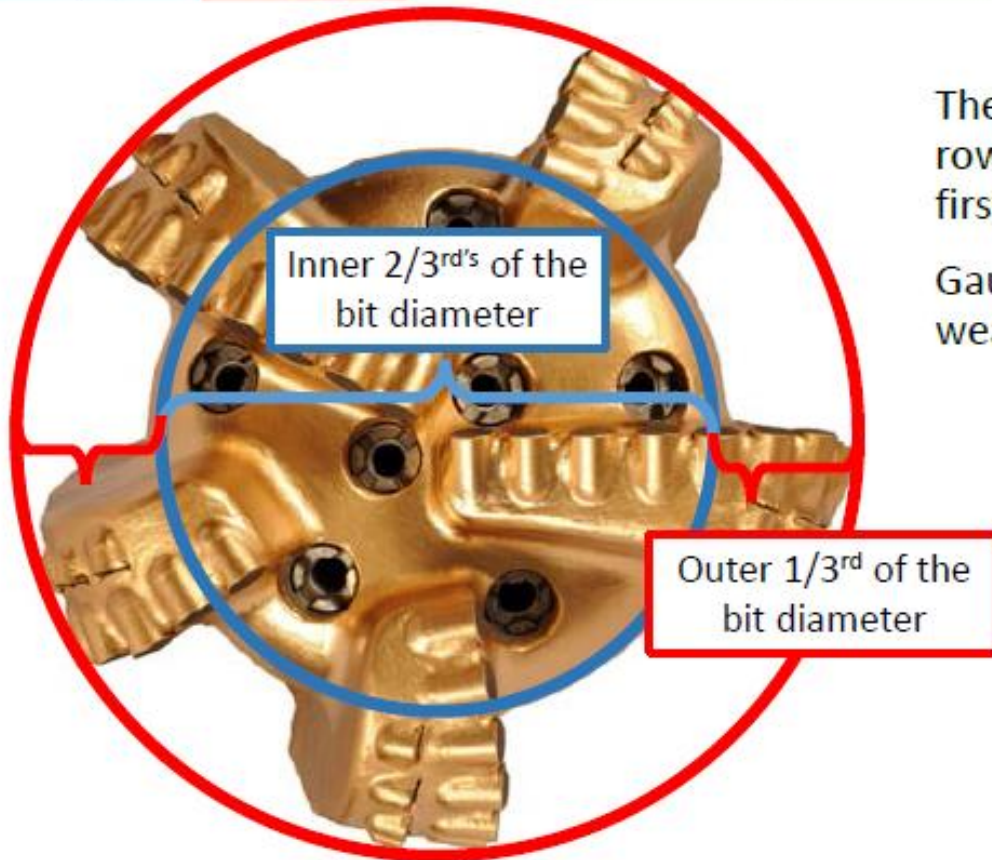


Inner Teeth 

Outer Teeth 

IADC Dull Grading for PDC

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Inner 2/3 ^{rd's})	Outer Rows (Outer 1/3 rd)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing Seals	Undersize (16 ^{th's} inch)	Other Dull Characteristic	Reason Pulled
Average Cutter Wear (0-8)	Average Cutter Wear (0-8)						



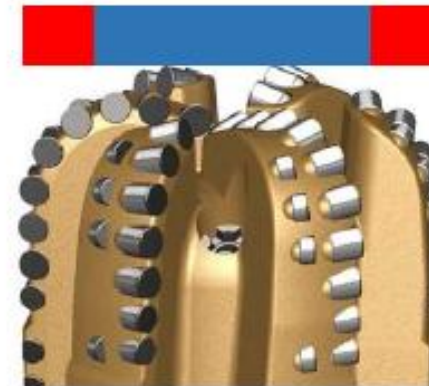
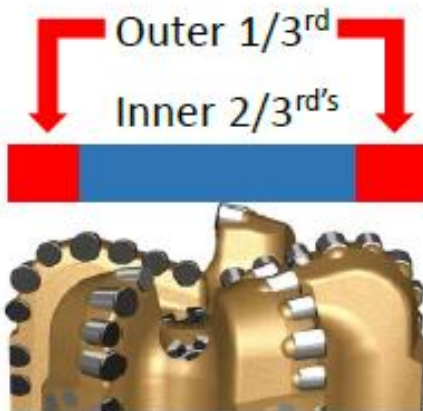
The average wear in the inner & outer rows are calculated and recorded in the first two boxes.

Gauge cutters are excluded from the wear in the outer rows

- Not 1/3 and 2/3 cutter count
- But radius

IADC Dull Grading for PDC

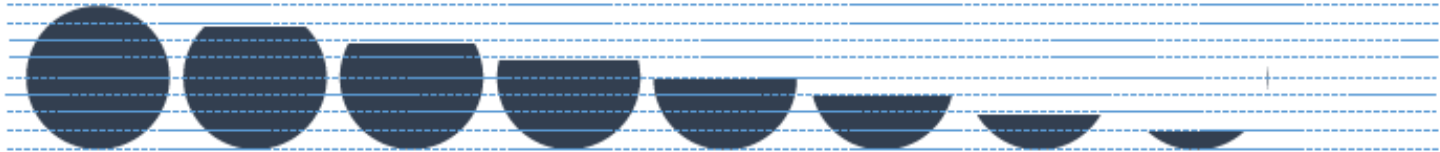
The highest cutter on each blade is the approximate transition between the Inner 2/3rd's & Outer 1/3rd



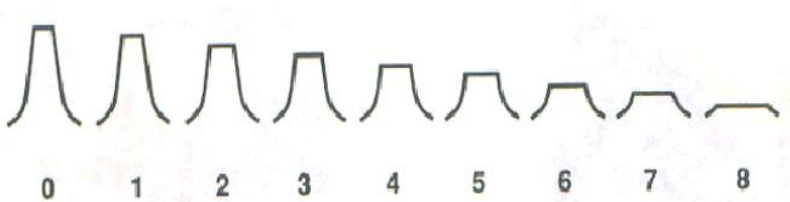
IADC Dull Grading

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Inner 2/3 rd 's)	Outer Rows (Outer 1/3 rd)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing Seals	Undersize (16 th 's inch)	Other Dull Characteristic	Reason Pulled
Average Cutter Wear	Average Cutter Wear						
Grade: 0-8	Grade: 0-8						

A grading scale between 0-8 is used to note the amount of wear in the Inner & Outer rows



Grade	0	1	2	3	4	5	6	7	8
Wear	0/8	1/8	2/8	3/8	4/8	5/8	6/8	7/8	8/8



Milltooth and Insert

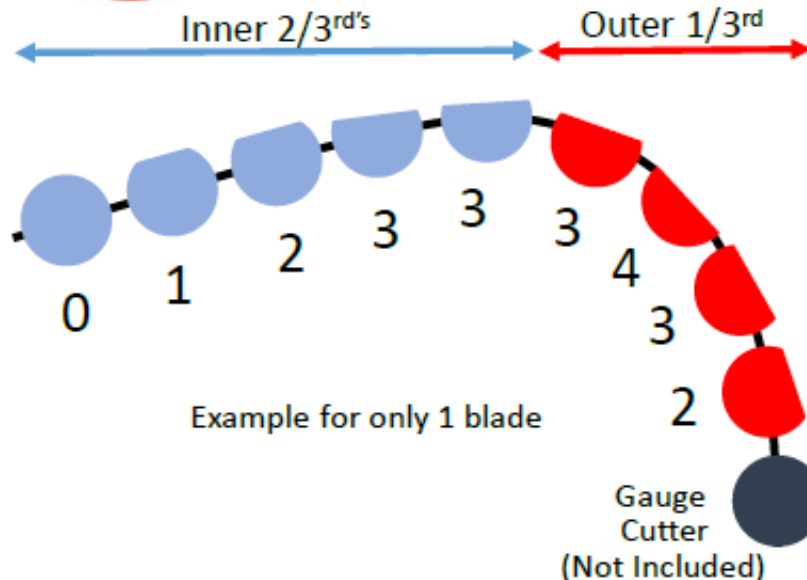
IADC Dull Grading for PDC

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Inner 2/3 rd 's)	Outer Rows (Outer 1/3 rd)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing Seals	Undersize (16 th 's inch)	Other Dull Characteristic	Reason Pulled
2	3						



Calculating Average Cutter Wear for Inner & Outer Rows

- 1) Grade each cutter with a 0 to 8
- 2) Add up the values
- 3) Divide by the number of cutters
- 4) Round to the nearest number



Example for only one Blade

Inner 2/3rd's Row

$$0 + 1 + 2 + 3 + 3 = 9$$

$$9 / 5 = 1.8 \text{ (round up)}$$

Inner 2/3rd's Wear = "2"

Outer 1/3rd Row

$$3 + 4 + 3 + 2 = 12$$

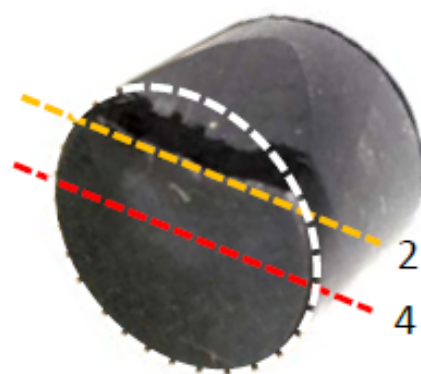
$$12 / 4 = 3$$

Outer 1/3rd Wear = "3"

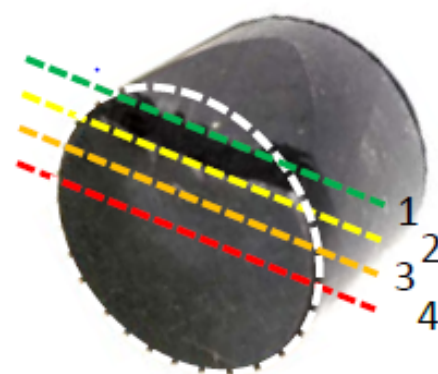
- The wear value (0-8) is the average wear condition, not the maximum amount of wear on one cutter.
- What is the minimum and maximum cutter wear?
 - The average will be between the minimum and maximum values
 - Is it closer to the minimum or maximum value?
- Establish the “4” cutter center line, 50% wear condition. Then estimate “2”, 25% wear condition. Does the cutter wear look closer to “1”, “2” or “3”. (See below)



Step 1)
Establish the center line, “4”



Step 2)
Estimate “2”, 50% from
center line to cutter tip

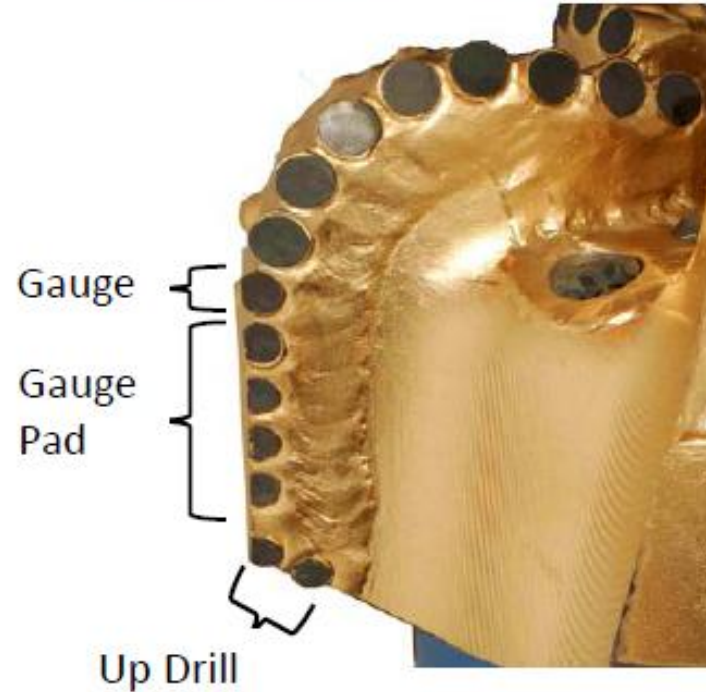


Step 3)
Is the wear closer to “1”,
“2” or “3”?

- Cutters with no wear are often incorrectly graded 1-1 to indicate the bit is used.
 - 0-0 is the correct grading



- Gauge, Gauge Pad & Up Drill cutters are not included in the Outer Row cutter wear calculation.



- Secondary row cutters are included in the Average Cutter Wear calculation.



- Cutter wear is measured across the diamond table and not the carbide substrate. If the diamond is missing and the carbide is intact, the cutter is graded as an "8".



Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows - Average Cutter Wear	Outer Rows – Average Cutter Wear	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing & Seals	Undersize (16 th 's inch)	Other Dull Characteristic	Reason Pulled
		What is the most notable cutter condition?					

Cutter	
BF	Bond Failure
BT	Broken Cutter
CT	Chipped Cutter
ER	Erosion
FC	Flat Crested Wear
HC	Heat Checking
LT	Lost Cutter
NO	No Characteristic
RG	Rounded Gauge
WT	Worn Cutters
	Unofficial:
DL	Delamination
SP	Spalled cutter

SPE 23939 – *Dull Characteristics* – “The most prominent or “primary” physical change from new condition of the cutter is recorded in the 3rd box.

“Other” dull characteristics of the bit are noted in the 7th box. The difference being that 3rd box describes cutter wear, while the 7th box may concern other wear characteristics of the bit as a whole.”

Most Dull Condition for PDC

Cutter

CT – Chipped Cutter

Diamond Edge Chipping



Cutter

BT – Broken Cutter

Fracture through the carbide substrate



Cutter

FC – Flat Crested Wear

Flat, Smooth Wear



Cutter

WT – Worn Cutter

Non Planar Wear or a combination of CT, BT, FC, ER



Most Dull Condition for Roller Cone

BC – Broken Cone



BT – Broken Teeth



CT – Chipped Teeth



LC – Lost Cone



WT – Worn Teeth



RG – Rounded Gage



LT – Lost Teeth



JD – Junk Damage



Dull Condition for PDC or Roller Cone

Bit Type	Code	Condition
Fixed Cutter or Roller Cone	BT	Broken Teeth/Cutters
	BU	Balled Up Bit
	CR	Cored
	CT	Chipped Teeth/Cutters
	ER	Erosion
	HC	Heat Checking
	JD	Junk Damage
	LN	Lost Nozzle
	LT	Lost Teeth/Cutters
	NO	No Dull Characteristics
	PN	Plugged Nozzle
	RR	Re-Runnable - Not recommended
	NR	Not Re-Runnable - Not recommended
	WO	Washed Out Bit
	WT	Worn Teeth/Cutters

Bit Type	Code	Condition
Roller Cone Only	BC	Broken Cone
	CC	Cracked Cone
	CD	Cone Dragged
	CI	Cone Interference
	FC	Flat Crested Wear
	LC	Lost Cone
	OC	Off Center Wear
	PB	Pinched Bit
	RG	Rounded Gage
	SD	Shirttail Damage
	SS	Self-Sharpening Wear
	TR	Tracking

Bit Type	Code	Condition
Fixed Cutter Only	BF	Bond Failure
	RO	Ring Out

Location

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Average Cutter Wear)	Outer Rows (Average Cutter Wear)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing & Seals	Undersize (16 th 's inch)	Other Dull Characteristic	Reason Pulled
			C, N, S, T, G, A				

Location of Primary Dull Characteristic

C	Cone
N	Nose
S	Shoulder
T	Taper
G	Gauge
A	All

More than one Location can be noted

G: Gauge

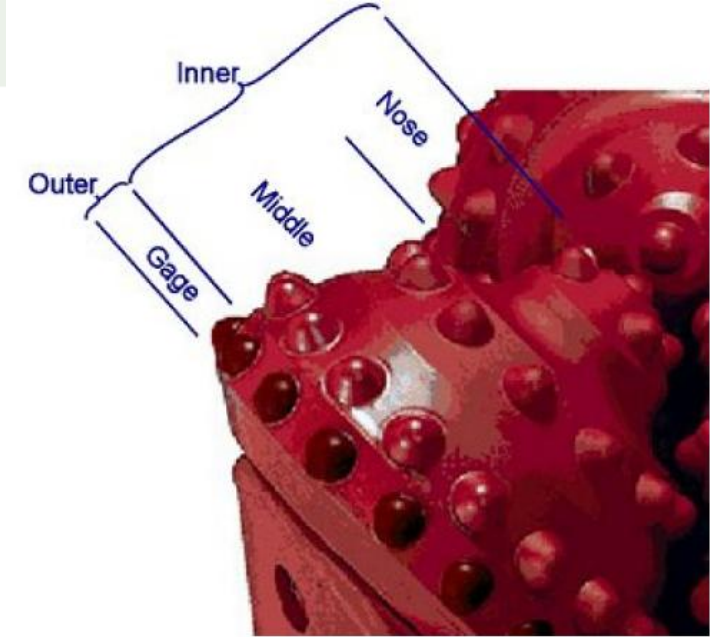
S: Shoulder – Curved

T: Taper - Straight

C: Cone

N: Nose

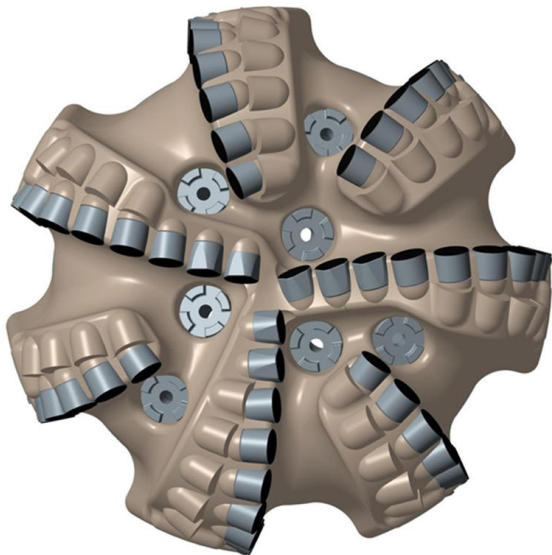
SPE 23939 – Location – “Used to indicate the location of the primary dull characteristic, 3rd box. One or more of these codes may be used to indicate the location.”



Bearing

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Average Cutter Wear)	Outer Rows (Average Cutter Wear)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing & Seals	Undersize (16 th 's inch)	Other Dull Characteristic	Reason Pulled
				X			

Fixed Cutter Bit



Roller Cone Bit



SPE 23939 – Bearing– “This box is used only for roller cone bits. It will always be marked “X” for fixed cutter bits

Gauge for PDC

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Average Cutter Wear)	Outer Rows (Average Cutter Wear)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing & Seals	Gauge Undersize	Other Dull Characteristic	Reason Pulled
					# 16 th Inch		

Gauge Undersize	
1	In Gauge
1	1/16 th undersize
2	2/16 th undersize
3	3/16 th undersize
4	4/16 th undersize
8	8/16 th undersize
...	...

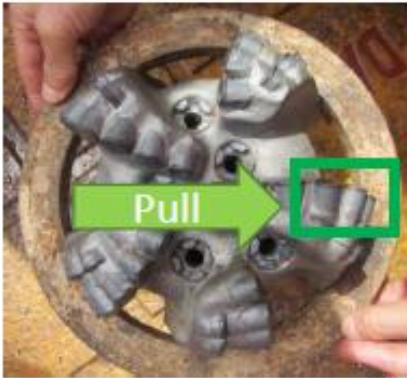


$1\frac{1}{2}'' = 24/16'' = \text{"24"}$

Procedure:

- 1) Use a "No-Go PDC" gauge ring
- 2) Place the gauge ring over the largest diameter: PDC cutter or Gauge Pad
- 3) Pull the ring tight to one side
- 4) Measure the opposing size
- 5) Record the distance to the nearest 1/16th of an inch

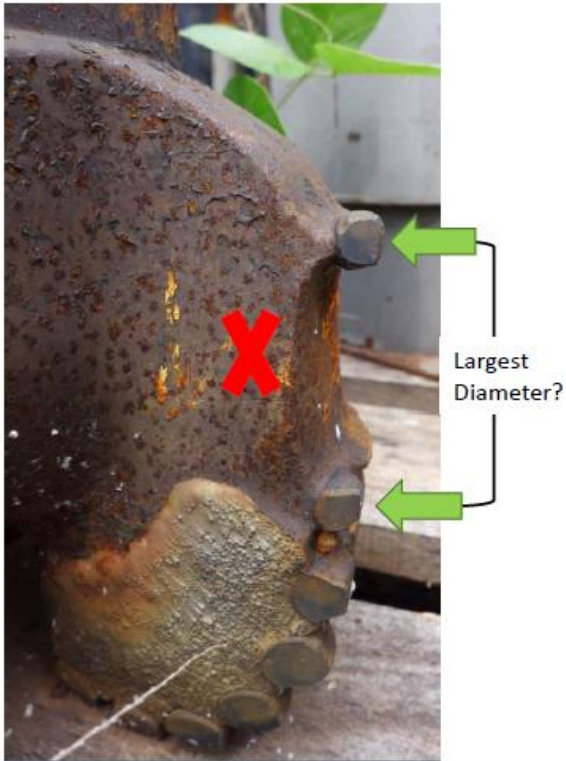
SPE 23939 – Gauge – "6th Box is used to record the condition of the bit gauge. "1" is used for the bit still in gauge. Otherwise, the amount the bit is under gauge is recorded to the nearest 1/16th of an inch"



Procedure:

No-Go PDC Gauge Ring

$2/16'' = \text{"2"}$

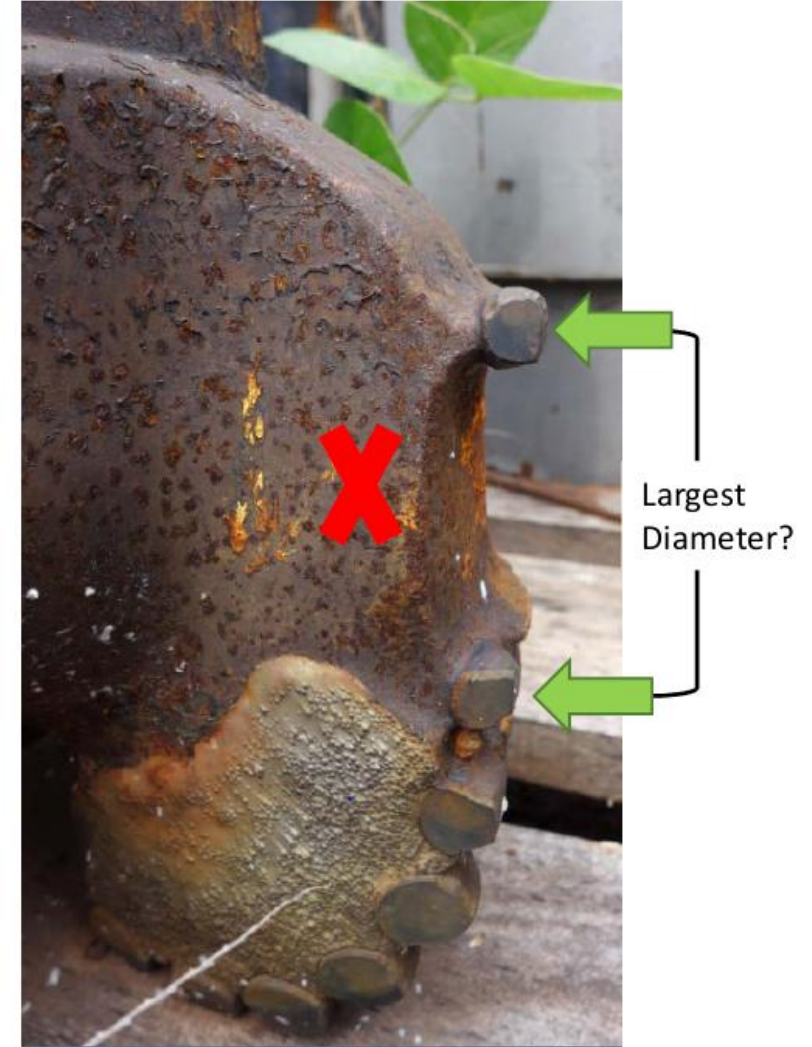
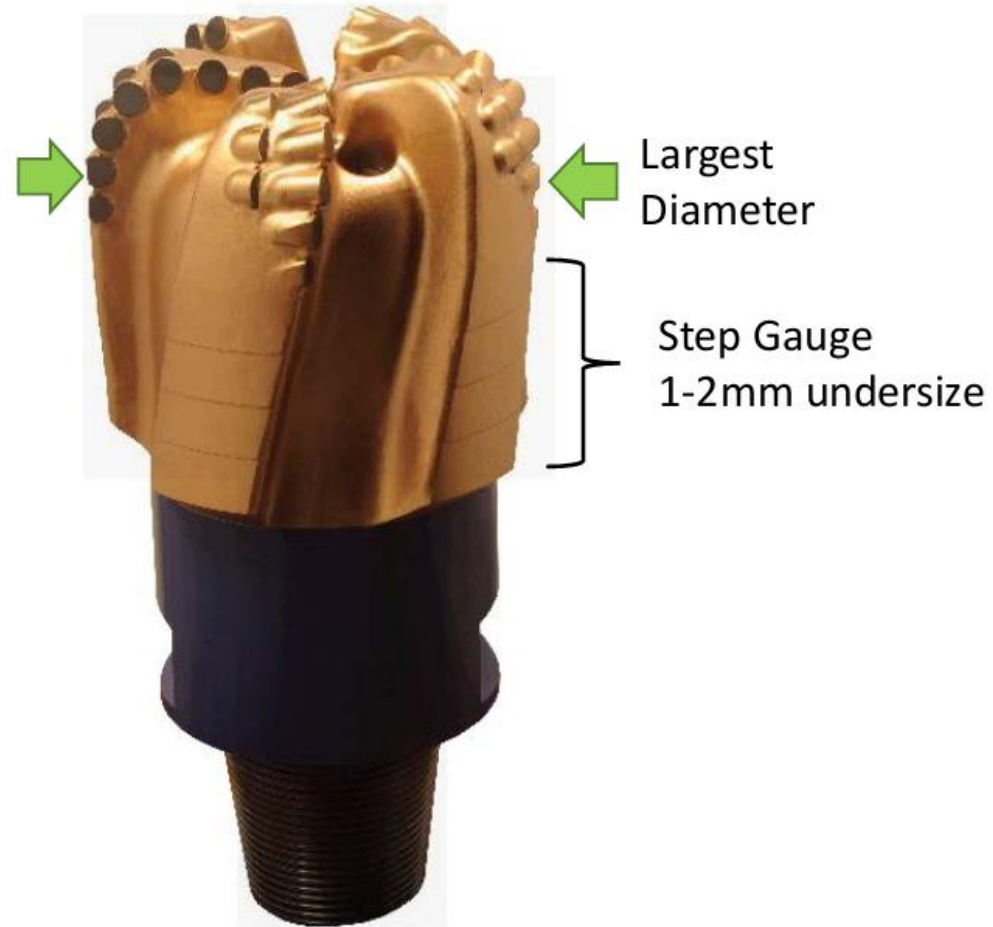


Gauge cutters are pre-ground during the manufacturing process. The amount of gauge cutter grind can vary by company and product line.



Undergauge

The gauge pad might be tapered or stepped. Thus, measure the largest diameter, typically the gauge cutters.



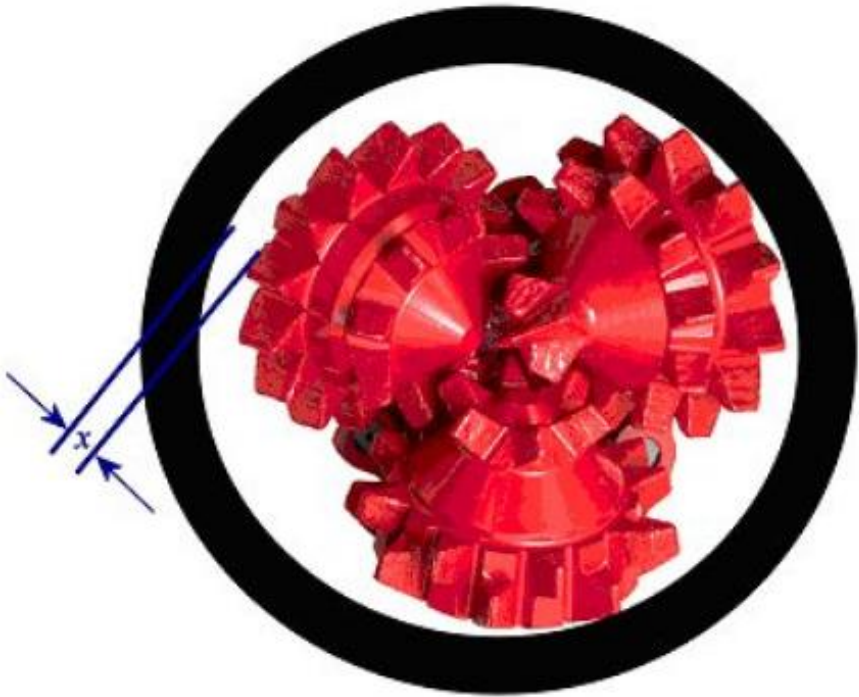
Gauge for Roller Cone

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Average Cutter Wear)	Outer Rows (Average Cutter Wear)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing & Seals	Gauge Undersize	Other Dull Characteristic	Reason Pulled
					# 16 th Inch		

Gauge Undersize	
I	In Gauge
1	1/16 th undersize
2	2/16 th undersize
3	3/16 th undersize
4	4/16 th undersize
8	8/16 th undersize
...	...

For instance, The “x” measurement is 3/16”, the actual undergauge is $3 \times \frac{2}{3} = \frac{2}{16}$ ”.

And a 2/16” undergauge would be listed as “2” in the dull grade.



$Gage = \frac{2}{3}x$

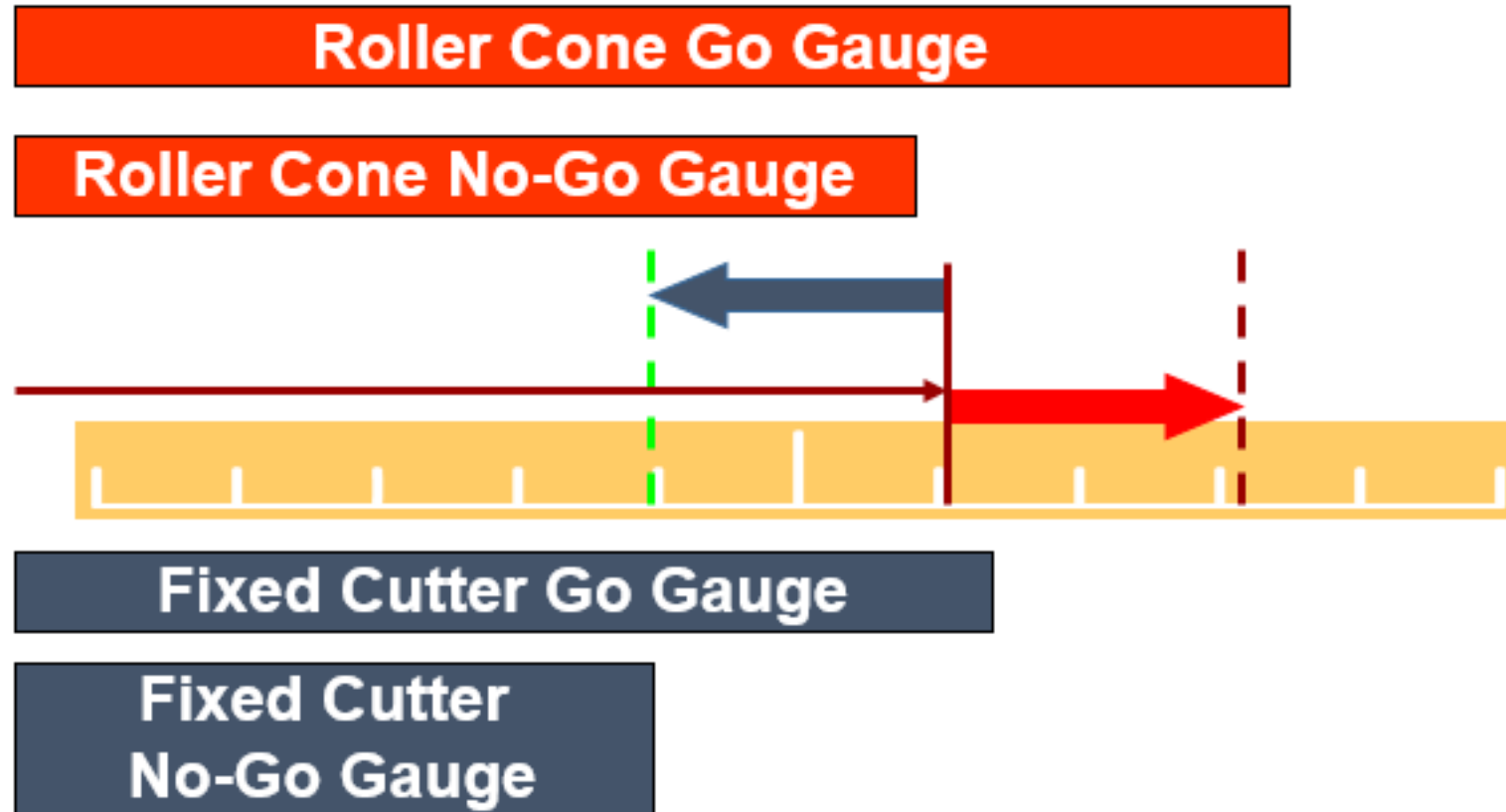
Gauge Measurement

- “I” In-Gauge is commonly mistaken as “1”, 1/16th.
 - Unofficially, “0” or “IN” are used in place of “I”.
- Use a “No-Go” gauge ring when dull grading PDC & Roller cones bits.
 - “No-Go” gauge ring: Measures minimum OD tolerance.
 - “Go” gauge ring: Measures maximum OD tolerance.
- PDC and Roller cone bits use different gauge rings.

It is not suggested to use PDC & Roller cone rings interchangeably.

 - PDC Bit tolerance: +0.000” to slightly undersized.
 - Roller Cone Bit tolerance: -0.000” to slightly oversized.

Gauge Tolerance



Gauge Tolerance

Roller Cone

Bit Size Range	Bit Tolerance	Go gauge	No-go gauge
3 3/8" – 13 3/4"	Nom. - 0 to + 1/32" (0.03125")	Nom Max + 0.03125" to + 0.03325"	Nom. - 0 to - 0.002"
14" – 17 1/2"	Nom. - 0 to + 1/16" (0.0625")	Nom Max + 0.0625" to + 0.0675"	Nom. - 0 to - 0.005"
17 5/8" +	Nom. - 0 to + 3/32" (0.09375")	Nom Max + 0.09375" to + 0.09875"	Nom. - 0 to - 0.005"

PDC

Bit Size Range	Bit Size Tolerance	Go gauge	No-go Gauge
Up to 6 3/4"	Nom. + 0 to -0.015	Nom. +.002 to -0.0	Nom. Min - 0.002 to - 0.005
6 25/32" to 9"	Nom. + 0 to -0.020	Nom. +.002 to -0.0	Nom. Min - 0.002 to - 0.005
9 1/32" to 13 3/4"	Nom. + 0 to -0.030	Nom. +.002 to -0.0	Nom. Min - 0.002 to - 0.005
13 25/32" to 17 1/2"	Nom. + 0 to -0.045	Nom. +.002 to -0.0	Nom. Min - 0.002 to - 0.005
17 17/32" +	Nom. + 0 to -0.063	Nom. +.002 to -0.0	Nom. Min - 0.002 to - 0.005

Other Dull Characteristic

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Average Cutter Wear)	Outer Rows (Average Cutter Wear)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing & Seals	Undersize (16 th s inch)	Other Dull Characteristic	Reason Pulled
		<i>(Only the cutter dull condition)</i>				What is the second most notable condition? Cutter & Bit Body	

Other Dull Characteristic

Cutter		Bit Body		Hydraulic		General	
BF	Bond Failure	CR	Cored	BU	Balled Up	NR	Not Rerunnable
BT	Broken Cutter	ER	Erosion	LN	Lost Nozzle	RR	Rerunnable
CT	Chipped Cutter	HC	Heat Checking	PN	Plugged Nozzle	NO	No Characteristics
ER	Erosion	JD	Junk Damage				
FC	Flat Crested Wear	RO	Ring Out				
HC	Heat Checking	WO	Wash Out				
LT	Lost Cutter						
NO	No Characteristic		Unofficial				
RG	Rounded Gauge	BB	Broken Blade				
WT	Worn Cutters						
	Unofficial:						
DL	Delamination						
SP	Spalled cutter						

SPE 23939 – *Dull Characteristics –*
“Secondary evidence of bit wear is noted. Such evidence may relate specifically to cutting structure wear, as recorded in the 3rd box, or may be identifiable wear of the bit as a whole, such as “erosion.” Many times this “secondary” dull grade identifies the cause of the dull characteristic noted in the 3rd Box.”

IADC Dull Grading for PDC

Bit Body

CR Cored

Loss of diamond in the center of the bit



IADC Dull Grading for PDC

Bit Body

RO Ring Out

Loss of diamond over a radial location, creating a “Ring” in the body.

Note: CR – Coring is cutter loss in the center of the bit



IADC Dull Grading for PDC

Bit Body

Irregular marks on the bit body, created by junk downhole

JD Junk Damage



Dull Grading – Reason Pulled

Cutting Structure				Bearing	Gauge	Remarks	
Inner Rows (Average Cutter Wear)	Outer Rows (Average Cutter Wear)	Primary Dull Characteristic	Location of Primary Dull Characteristic	Bearing Seals	Undersize (16 th 's inch)	Other Dull Characteristic	Reason Pulled

Reason Pulled	
BHA	Changed Bottom Hole Assembly
DMF	Downhole Motor Failure
DSF	Drill String Failure
DST	Drill Stem Test
DTF	Downhole Tool Failure
LOG	Run Logs
RIG	Rig Repair
CM	Condition Mud
CP	Core Point
DP	Drill Plug
FM	Formation Change
HP	Hole Problems
PP	Pump Pressure
PR	Penetration Rate
TD	Total Depth
TQ	Torque
TW	Twist-Off
WC	Weather Conditions
WO	Washout – Drill String

PDC - Normal Dull Conditions

NO = No Dull



WT = Worn Teeth



Impact Damage

ReedHycalog

NOV Wellbore
Technologies

CT – Chipped Teeth



BT = Broken Teeth



Severe Damage

HC = Heat Checking



HC = Heat Checking



Heat checking to matrix body:

Stick slip

RO = Ring Out



CR = Cored



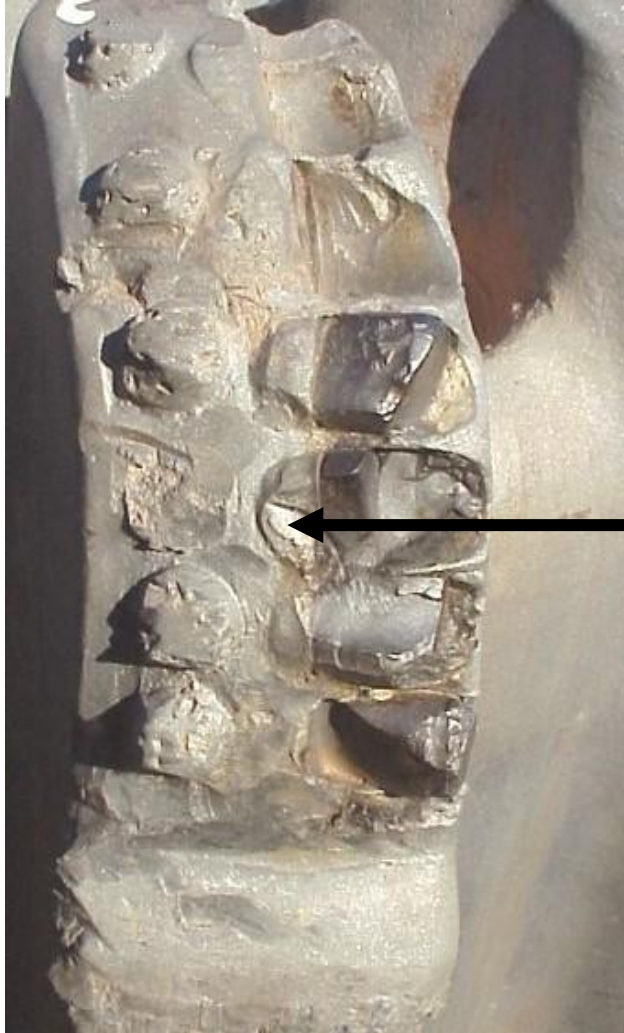
LN = Lost Nozzle



No Code – Broken Blade



JD – Junk Damage



Massive catastrophic damage to the cutting structure and/or gauge pads and/or bit body. Check for rust marks on Matrix bodied bits.

Hydraulic Related Conditions

ER = Erosion



PN = Plugged Nozzle



BU = Balled Up



Quality Control

LT = Lost Teeth



BF- Bond Failure





Questions?